SIMATS ENGINEERING

Department of Information Technology

ITA05 – Computer Vision

ASSIGNMENT 1 – CO1

Apply, Compare and Justify Image Acquisition, Formation and Processing Techniques for a Textile Fabric Defect Inspection System

CO1: Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems.

Particulars	**Details**	**Particulars**	**Details**
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Industrial Case Study: Sri Lakshmi Textile Industries, Tiruppur

INTRODUCTION

1. REPORT BASIS, ASSUMPTIONS AND EVIDENCE POLICY

This report presents a systematic study of the application, comparison, and justification of image acquisition, image formation, and image processing techniques for developing a computer vision-based textile fabric defect inspection system for Sri Lakshmi Textile Industries, Tiruppur. The primary purpose of the proposed system is to identify visible defects in textile fabrics accurately and consistently while the fabric moves through an industrial inspection line. In conventional textile inspection, defects such as holes, stains, broken yarns, missing threads, uneven weaving patterns, and surface irregularities may be identified through manual observation. However, manual inspection can be affected by operator fatigue, differences in individual judgement, changing environmental conditions, and the speed of the production process. A computer vision system can help overcome these limitations by acquiring digital images of the fabric, analyzing the visual information contained in those images, and identifying regions that differ from the expected fabric appearance. The report therefore examines the complete process, beginning with the selection of an appropriate image acquisition arrangement and continuing through image formation, preprocessing, feature extraction, defect detection, performance evaluation, and final inspection decisions.

The technical discussion in this report is based on the industrial scenario and engineering requirements defined for the proposed textile inspection system. The report considers a fabric width of approximately 1000 mm, a camera field of view of 1200 mm × 400 mm, and a fabric movement speed of 0.6 m/s under the initial operating conditions. The proposed system is expected to detect defects with a minimum target size of approximately 1.5 mm, while maintaining a fabric-width measurement accuracy of ±2 mm and producing an inspection decision within 0.5 seconds. Additional design considerations include variations in lighting, low-contrast oil stains, camera mounting arrangements, processing hardware, image storage, and the need to integrate the inspection system into an existing textile production environment. These values are treated as design requirements and engineering assumptions for analyzing possible solutions; they should not be interpreted as experimentally verified performance results. Actual performance would need to be established through practical implementation, controlled experiments, representative fabric samples, and validation under real industrial operating conditions.

The report follows an evidence-based approach by distinguishing established computer vision concepts, proposed design decisions, engineering assumptions, and results that would require experimental verification. Fundamental concepts such as image sensing, image formation, sampling, quantization,

illumination, grayscale conversion, filtering, thresholding, edge detection, texture analysis, and morphological processing are used to explain how the proposed system may operate. Where a technique is recommended, the justification is based on its expected suitability for the stated inspection problem, including defect visibility, computational cost, processing speed, and implementation complexity. For example, grayscale conversion may simplify intensity-based analysis, while contrast enhancement may help make faint stains more distinguishable from the surrounding fabric. Similarly, morphological operations may assist in removing small isolated responses or refining detected defect regions. Nevertheless, no single technique is assumed to be effective for every type of fabric defect. A method that works well for a dark stain on a light fabric may perform poorly when detecting a small hole, a broken yarn, or a defect that has nearly the same intensity as the surrounding material. Consequently, the report emphasizes the need to compare alternative techniques and justify their selection according to the specific characteristics of the inspection task.

The evidence policy also requires the limitations of the proposed system to be stated clearly. Computer vision performance can be affected by fabric movement, vibration, camera resolution, lens distortion, uneven illumination, shadows, surface texture, and differences between fabric colours or patterns. A detection method may also produce false positives when normal variations in weaving are incorrectly identified as defects, or false negatives when genuine defects are too small, faint, or visually similar to the background. Therefore, the proposed system must be evaluated using suitable performance measures, including detection accuracy, precision, recall, false-positive rate, missed-defect rate, processing time, and measurement error, where appropriate. Any numerical performance values reported in a final experimental study should be supported by recorded test results rather than assumptions. If actual industrial testing has not been completed, the report should describe the values as targets, estimates, or proposed acceptance criteria. This distinction ensures that the analysis remains technically responsible and provides a reliable foundation for further development, implementation, and testing of the textile fabric defect inspection system.

2. INDUSTRIAL SCENARIO AND SHARED DESIGN DATA POLICY

Sri Lakshmi Textile Industries, located in Tiruppur, is used as the industrial case study for examining the application of computer vision to fabric quality inspection. The proposed inspection system is intended to examine fabric continuously as it passes through a production line and to identify visible defects before the material proceeds to later manufacturing, finishing, packing, or dispatch operations. Textile defects may originate from several sources, including irregular yarn tension, broken threads, weaving faults, contamination, oil leakage, mechanical problems, and inconsistencies in the production process. If such

defects are not identified at an appropriate stage, they may reduce the quality of the finished material, increase material wastage, require additional rework, and affect customer satisfaction. A camera-based inspection arrangement can provide a consistent method of capturing fabric images and applying predefined processing rules to identify suspicious regions. The proposed system is not intended to replace every aspect of human quality control; rather, it is designed to support inspectors by providing repeatable visual analysis, recording detected defect locations, and helping operators make timely decisions about the inspected material.

For the purpose of comparing image acquisition and processing techniques, the report uses a shared set of design data so that different methods can be assessed under the same operating conditions. The fabric width is assumed to be 1000 mm, while the camera field of view is specified as 1200 mm × 400 mm to provide coverage across the fabric width and a section of its length. The initial fabric movement speed is set at 0.6 m/s, and the target minimum detectable defect size is 1.5 mm. These parameters influence the choice of camera resolution, lens, exposure time, illumination arrangement, image sampling density, and processing strategy. For example, the selected imaging system must provide enough spatial detail to represent a defect of the specified size using a sufficient number of pixels. The exposure time must also be short enough to limit motion blur as the fabric moves through the field of view. If the camera captures images too slowly or the exposure time is too long, a defect may become blurred or may not be represented clearly enough for reliable detection. Similarly, the field of view must be large enough to cover the required fabric area without sacrificing excessive image detail.

Lighting is another important component of the shared design requirements because the appearance of a fabric defect depends not only on the defect itself but also on the illumination and surface properties of the material. The initial scenario considers environmental illumination that may vary between approximately 500 and 3000 lux, as well as oil stains that may have only a small grayscale contrast of approximately 5–8 intensity levels against the surrounding fabric. These conditions indicate that the inspection system should not depend entirely on a fixed intensity threshold without considering the possibility of uneven lighting and low-contrast defects. A controlled lighting arrangement, appropriate exposure settings, and suitable image enhancement techniques may help improve the consistency of the captured images. The camera is assumed to be mounted approximately 700 mm above the fabric, and the distance between the inspection camera and the marking mechanism is assumed to be 0.3 m. These distances are relevant when estimating the physical area represented by each image, determining the location of detected defects, and coordinating the inspection output with a marking or rejection

mechanism. The final installation would require practical measurement and calibration to confirm that these assumptions are appropriate for the actual production line.

The proposed processing arrangement is based on a computer with a four-core processor and 16 GB of RAM, using Python, OpenCV, and NumPy for image acquisition support, image manipulation, analysis, and algorithm development. Gigabit Ethernet is considered as the image-transfer connection, while a 2 TB storage device is proposed for retaining inspection records and selected image data. The initial budget is set at approximately ₹150,000, and a seven-day retention period is considered for stored inspection information. These values establish a practical framework for comparing possible solutions without assuming that every component has already been purchased, installed, or tested. The design must balance image quality, computational demand, storage requirements, installation complexity, and cost. A high-resolution camera may improve the visibility of small defects, but it can also increase image size, transfer time, memory consumption, and processing load. Similarly, a computationally intensive image analysis method may identify complex defect patterns but may not satisfy the required processing deadline on the specified hardware. The selection of the final system must therefore consider the complete inspection pipeline rather than evaluating the camera, illumination, processing algorithm, or computer in isolation.

The shared design data policy also establishes a common basis for comparing alternative techniques. For example, conventional thresholding, adaptive thresholding, filtering, edge detection, texture analysis, and morphology-based processing should be assessed using comparable fabric images and equivalent operating conditions wherever possible. The comparison should consider not only whether a method detects a defect but also whether it incorrectly identifies normal fabric patterns as defects, how much processing time it requires, and whether it can handle changes in illumination or fabric appearance. The target missed-defect rate is set at no more than 2%, but this is an intended acceptance criterion rather than a confirmed result. It must be verified using a representative test dataset containing both defective and defect-free fabric samples. The shared design data therefore provides a consistent basis for engineering decisions, while the actual effectiveness of each method remains dependent on implementation and experimental validation.

3. EXPERIMENTAL SETUP AND ENGINEERING ASSUMPTIONS

The proposed experimental setup consists of a fabric movement arrangement, an image acquisition unit, a controlled illumination system, an image processing computer, and an inspection output mechanism. The fabric is assumed to move through the inspection area at a relatively constant speed of 0.6 m/s. A camera positioned above the fabric captures images covering the required inspection region, while

suitable lighting helps reveal changes in colour, texture, shape, and surface continuity. The camera and lens must be selected according to the field of view, working distance, required spatial resolution, and expected defect dimensions. Before conducting the inspection, the imaging arrangement should be calibrated so that image coordinates can be related to physical distances on the fabric. This calibration is important when estimating the width of a defect, determining its location, or connecting a detected region to a marking mechanism. The camera should also be mounted securely to reduce image variation caused by vibration, and the fabric should be guided as consistently as possible to limit changes in position and focus during image acquisition.

Image acquisition is the first technical stage of the experiment because all later processing depends on the quality of the captured image. The experimental procedure should examine whether the camera can represent the complete fabric width, capture sufficient detail for the minimum target defect, and produce images with limited motion blur. The camera resolution must be considered together with the physical field of view rather than being evaluated independently. If the width of the inspection region is represented by too few pixels, a small defect may occupy only one or two pixels and become difficult to distinguish from noise or ordinary fabric texture. Conversely, an unnecessarily large image may increase processing time and memory requirements without producing a useful improvement in detection. The exposure time, frame rate, lens characteristics, camera orientation, and illumination direction must therefore be chosen together. The experiment should also consider whether a conventional area-scan camera is adequate or whether a line-scan camera would provide advantages for continuous fabric inspection. This decision depends on the movement of the material, the required image dimensions, the available installation space, and the speed at which the system must inspect the fabric.

The lighting arrangement should be evaluated under controlled and repeatable conditions. Although the initial design considers environmental illumination between approximately 500 and 3000 lux, relying on ambient light alone may introduce unwanted variation between images. A dedicated lighting arrangement may provide more consistent illumination and improve the visibility of low-contrast defects, including oil stains that differ from the surrounding fabric by only approximately 5–8 grayscale intensity levels. The experiment should compare the appearance of representative defects under different suitable lighting positions and intensities, while avoiding excessive reflections, deep shadows, and uneven brightness. Diffuse lighting may be useful when reducing harsh reflections, whereas directional lighting may help reveal raised fibres, holes, wrinkles, or surface irregularities through their effect on light and shadow. The most appropriate arrangement must be selected according to the fabric material, colour, weave

pattern, and defect type. Images should be collected under both normal and challenging conditions so that the system can be evaluated for robustness rather than only under ideal laboratory illumination.

After image acquisition, the captured images are processed using a sequence of computer vision operations. The initial stages may include image validation, resizing where appropriate, grayscale conversion, contrast adjustment, and noise reduction. These operations should be selected carefully because excessive smoothing or aggressive enhancement can remove important defect information or create artificial patterns that interfere with detection. Thresholding may then be used to separate regions with unusual intensity from the background, while edge detection may help identify discontinuities in yarn structure or fabric boundaries. Texture-based methods can be investigated when defects appear as changes in repeated weaving patterns, and morphological operations may help refine the detected regions by removing small isolated responses or connecting fragmented components. The output of the processing pipeline should identify suspicious regions and, where possible, estimate their position and dimensions. The system should retain enough intermediate information to support debugging and performance analysis, especially when a defect is missed or a normal fabric feature is incorrectly classified. The experimental implementation may be developed using Python, OpenCV, and NumPy on the specified four-core computer with 16 GB RAM, subject to confirmation that the selected processing method can meet the required timing constraints.

The experimental dataset should contain representative images of defect-free fabric and fabric with different types of defects, including stains, holes, broken yarns, missing threads, and irregular weaving patterns. Each image should be labelled according to the known condition of the inspected fabric so that the system's output can be compared with the reference result. The dataset should include variations in fabric colour, texture, lighting, defect size, and defect location wherever suitable samples are available. The data should be divided into development and evaluation sets when a learning-based method is used, so that the final performance is not assessed only on images already used to tune the model. For a conventional image processing method, the same principle of using separate evaluation samples helps determine whether the selected thresholds and processing parameters generalize to new fabric images. Performance should be measured using suitable indicators such as precision, recall, false-positive rate, missed-defect rate, processing time, and defect-size measurement error. The target of detecting defects as small as 1.5 mm and maintaining a missed-defect rate of no more than 2% should be treated as a requirement to be tested rather than a result that can be assumed in advance.

The engineering assumptions must also account for changes in production requirements. In the initial operating scenario, the system is expected to inspect fabric moving at 0.6 m/s and produce a decision

within 0.5 seconds. A separate changed-export-order scenario considers defects as small as 0.4 mm, possible ultraviolet-visible stains, fabric movement at 1.2 m/s, twice the original inspection volume, and a decision deadline of 0.3 seconds, while retaining the existing computer and budget constraints. These changes create additional challenges for image resolution, illumination, data transfer, processing speed, and detection reliability. A defect that can be represented adequately at the initial operating speed may become more difficult to detect when its size decreases and the fabric moves faster. Ultraviolet or other specialized illumination may also require compatible lighting and imaging equipment rather than simply changing the image processing software. The revised requirements should therefore be evaluated separately, and any claim that the existing setup can satisfy them must be supported by calculation and experimental testing. If the available hardware cannot meet the revised requirements, the report should identify the bottleneck and recommend feasible changes to the acquisition arrangement, processing pipeline, or operating procedure. This approach ensures that the proposed textile fabric defect inspection system is evaluated as a complete engineering solution, with its capabilities, limitations, assumptions, and future i

TASK 1: OVERVIEW AND FEATURES OF COMPUTER VISION – COMPARE GRADING/INSPECTION APPROACHES

1. Working Principle of Each Inspection Approach

A. Manual Inspection by Trained Workers

Manual inspection is a traditional fabric quality inspection method in which trained workers visually examine the fabric to identify defects such as holes, oil stains, broken yarns, missing threads, uneven weaving patterns, and colour variations. The fabric is spread or moved across an inspection table under suitable lighting, allowing the worker to observe its surface carefully. When a defect is identified, the worker may mark its location, record its type, and decide whether the affected fabric should be accepted, repaired, or rejected. The accuracy of this approach depends on the worker's experience, eyesight, concentration, lighting conditions, and ability to inspect the fabric at the required production speed. Manual inspection is relatively easy to establish because it does not require cameras, image processing software, or specialized computing equipment. However, inspecting large quantities of fabric continuously can be tiring, and small or low-contrast defects may be overlooked, especially during long working hours.

B. Image-Processing-Assisted Inspection

Image-processing-assisted inspection combines camera-based image acquisition and computer image processing with human decision-making. In this approach, a camera captures images of the moving fabric, and computer vision algorithms process the images to highlight possible defects. Techniques such as grayscale conversion, contrast enhancement, filtering, thresholding, edge detection, and texture analysis may be used to identify unusual regions in the fabric. The processed images, detected regions, or warning indicators are displayed to an operator, who examines the results and decides whether a suspected region represents an actual defect. This approach reduces the amount of fabric that the operator must examine entirely through unaided visual observation because the software directs attention towards suspicious regions. It also combines the speed and repeatability of computer processing with the experience and judgement of a trained worker. However, its overall performance still depends on the operator's ability to review the flagged regions correctly and quickly. If the software generates too many false alarms, the operator may experience additional workload and inspection delays.

C. Fully Automated Computer Vision Inspection

Fully automated computer vision inspection uses cameras, suitable illumination, image processing algorithms, and decision-making software to inspect fabric with minimal or no routine human intervention. The camera continuously captures images of the fabric as it moves through the production line. The system processes the images, identifies potential defects, classifies them according to predefined rules or a trained detection model, and generates an inspection result. Depending on the installation, it can record defect coordinates, estimate defect dimensions, store images, generate quality reports, and activate an alarm or marking mechanism when a defect is detected. Unlike manual inspection, the system does not become tired or lose concentration during continuous operation. It can also apply the same detection rules repeatedly, improving consistency between inspections. Nevertheless, the system must be properly calibrated and validated because performance may be affected by poor illumination, motion blur, unusual fabric patterns, changes in material colour, and defects that are not adequately represented in its detection method. Human supervision remains valuable for reviewing uncertain results, maintaining the equipment, and updating the system when production conditions change.

2. Comparison of the Three Inspection Approaches

The three approaches differ in their ability to identify defects, process fabric continuously, maintain consistent inspection standards, and support industrial production. Manual inspection relies mainly on human visual judgement, image-processing-assisted inspection combines software-generated results with operator verification, and fully automated inspection depends primarily on a computer vision system to produce decisions. The comparison below considers their suitability for the operating requirements of Sri Lakshmi Textile Industries.

Comparison factor	Manual inspection	Image-processing-assisted inspection	Fully automated computer vision
Working principle	Workers visually examine fabric and identify defects.	Software highlights suspicious regions for an operator to verify.	Cameras and algorithms detect and report defects automatically.
Accuracy	Depends on worker experience, eyesight, concentration, and lighting. Small defects may be missed.	Can improve defect identification by combining image analysis with human judgement.	Can provide reliable detection when properly designed, calibrated, and validated. Performance depends on the algorithm and image quality.

Comparison factor	Manual inspection	Image-processing-assisted inspection	Fully automated computer vision
Speed	Usually limited by human observation and the need for careful examination.	Faster than unaided inspection for locating suspicious regions, but operator verification takes time.	Can process images rapidly and support continuous inspection if hardware and software meet production requirements.
Consistency	May vary between workers and during long inspection periods.	More consistent than purely manual inspection, although final decisions may vary between operators.	Applies predefined processing rules consistently, provided operating conditions remain within the validated range.
Labour requirement	Requires workers to examine the fabric directly.	Requires both technical equipment and an operator to review results.	Requires fewer workers for routine inspection but still needs supervision, maintenance, and quality verification.
Continuous production suitability	Less suitable for very high-speed production over extended periods because of fatigue and attention limitations.	Suitable for production lines where operators can review software-generated results at the required speed.	Highly suitable for continuous production when image acquisition, processing speed, and detection performance meet the production requirements.
Initial investment	Generally low because specialized equipment is not essential.	Moderate because cameras, lighting, software, and computing equipment are required.	Usually higher because reliable acquisition hardware, processing equipment, integration, and validation are required.
Main limitation	Fatigue, subjective judgement, and possible missed defects.	Dependence on operator review and possible false alarms from the software.	Sensitivity to changes in lighting, fabric appearance, calibration, and algorithm capability.

The comparison indicates that manual inspection may be appropriate for small production volumes, occasional quality checks, or situations in which specialized equipment is not available. Image-processing-assisted inspection provides a practical intermediate solution because the computer helps identify suspicious areas while the operator retains control over the final decision. Fully automated computer vision offers the strongest potential for continuous, repeatable inspection because images can be analyzed systematically without requiring an operator to visually examine every section of fabric.

However, automation does not automatically guarantee perfect accuracy. The system's performance must be measured using representative fabric samples and verified under realistic production conditions.

3. Selection of the Most Suitable Approach for Sri Lakshmi Textile Industries

The most suitable approach for Sri Lakshmi Textile Industries is fully automated computer vision inspection, supported by periodic human supervision and manual verification of uncertain or flagged defects.

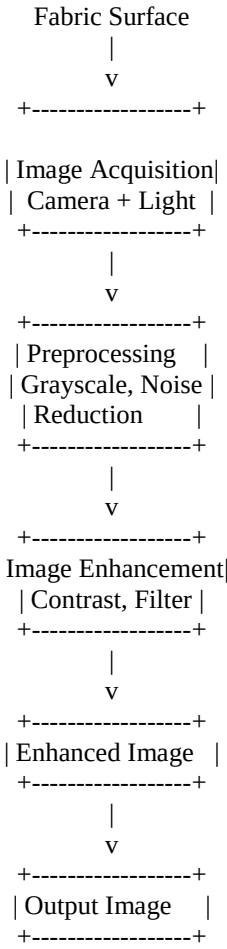
TASK 2: LEVELS OF COMPUTER VISION – PIPELINE DESIGN AND OPERATION

1. Introduction

A computer vision pipeline is a sequence of processing stages that converts an acquired image into meaningful information or a final decision. In textile fabric defect inspection, the pipeline receives images of moving fabric and analyzes them to identify defects such as holes, broken threads, oil stains, and weaving irregularities. The design of the pipeline determines how image information is processed, how defects are represented, how decisions are generated, and how quickly the system can respond. Different pipeline designs offer different advantages in terms of computational complexity, explainability, hardware requirements, detection capability, and response time.

2. Pipeline 1: Low-Level Processing Pipeline

2.1 Block Diagram



2.2 Working Principle

A low-level processing pipeline concentrates on acquiring an image and improving its visual quality through basic image processing operations. The camera captures the fabric surface under suitable illumination, and the captured image is passed through preprocessing operations to prepare it for further analysis. These operations may include resizing, grayscale conversion, noise reduction, and brightness normalization. Image enhancement techniques, such as contrast stretching and histogram equalization, may then be applied to improve the visibility of important image details. The final output is an enhanced image in which certain fabric characteristics or defects may become easier to observe.

For example, contrast enhancement may make an oil stain more visible when it has only a small intensity difference from the surrounding fabric. Noise reduction may suppress unwanted variations caused by the camera sensor or environmental conditions. However, a low-level pipeline primarily improves the image rather than independently understanding the meaning of the detected regions. It does not necessarily distinguish a genuine hole from an ordinary dark region unless additional detection logic is included. Therefore, this pipeline is useful as a foundation for other computer vision systems, but by itself it is insufficient for reliable identification and classification of all the required fabric defects.

2.3 Input and Output of Each Stage

Stage 1: Image Acquisition

The input to image acquisition is the physical fabric surface moving through the inspection area. The camera, lens, and lighting arrangement capture the reflected light and convert it into a digital image. The output is a raw image representing the fabric surface. The image quality depends on camera resolution, exposure time, focus, illumination, and fabric movement. For Sri Lakshmi Textile Industries, the acquisition arrangement must cover the required.

Stage 2: Preprocessing

The input is the raw image obtained from the camera. Preprocessing prepares the image for subsequent operations by reducing unwanted noise, converting colour information into grayscale when appropriate, and adjusting the image dimensions if necessary. Grayscale conversion represents the image using intensity values instead of separate colour channels, which can reduce computational requirements when colour information is not essential. The output is a prepared image with a more convenient representation for further processing.

Stage 3: Image Enhancement

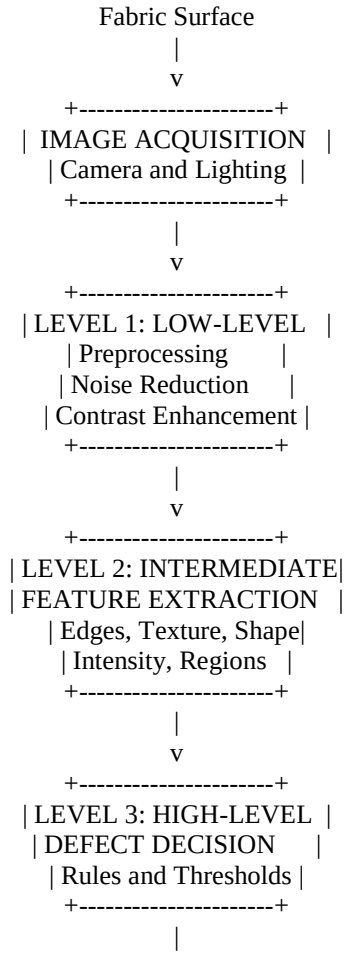
The input is the preprocessed image. Enhancement operations improve the visibility of relevant details by adjusting contrast, brightness, or local intensity differences. Suitable filtering may also be applied to suppress noise while preserving useful edges. The output is an enhanced image that may make defects easier to inspect or process. Enhancement must be selected carefully because excessive filtering can remove thin broken threads or alter the appearance of small holes.

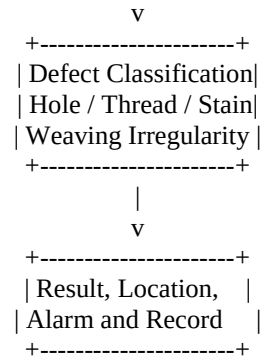
Stage 4: Enhanced Image Output

The input is the enhanced image produced by the previous stage. The system displays or stores the image for later examination or further processing. The output is an improved image, not necessarily a final defect classification. An additional detection stage would be required to determine the type, location, and severity of each defect.

3. Pipeline 2: Three-Level Computer Vision Pipeline

3.1 Block Diagram





3.2 Working Principle

The three-level pipeline extends basic image processing by adding intermediate-level feature extraction and high-level decision-making. Its first level acquires and prepares the fabric image using operations such as grayscale conversion, noise reduction, contrast enhancement, and illumination correction. The second level extracts meaningful features from the prepared image. These features may include edges, intensity differences, texture patterns, connected regions, shape characteristics, and deviations from the expected fabric structure. The third level uses the extracted features and predefined decision rules to determine whether a region represents a defect and, where possible, classify it into a defect category.

3.3 Input and Output of Each Stage

Stage 1: Image Acquisition

The input is the fabric surface under the camera's field of view. The camera captures a digital image using suitable illumination and exposure settings. The output is a raw fabric image containing the visual information needed for defect analysis.

Stage 2: Low-Level Processing

The input is the raw image. The system performs preprocessing operations such as grayscale conversion, noise reduction, brightness normalization, and contrast enhancement. The purpose is to reduce unwanted image variation and improve the visibility of defect-related information. The output is a prepared image suitable for feature extraction.

Stage 3: Intermediate-Level Feature Extraction

The input is the preprocessed image. The system calculates or identifies useful image features, including intensity variations, edges, shapes, connected components, and texture characteristics. Edge detection

may help identify broken lines or thread discontinuities. Shape and region analysis may assist in detecting holes, while intensity analysis may help locate stains. Texture analysis may identify deviations from the expected weaving pattern. The output is a set of numerical measurements, feature maps, or candidate defect regions that summarize the important visual characteristics of the fabric.

Stage 4: High-Level Defect Decision

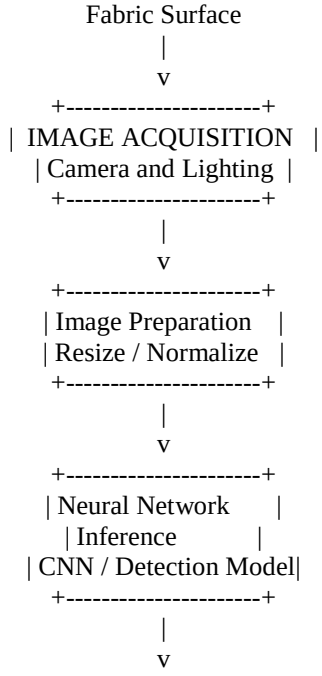
The input is the extracted feature information. The decision stage applies predefined rules, thresholds, or a lightweight classifier to determine whether a candidate region is normal or defective. If sufficient features are available, the system may classify the defect as a hole, broken thread, stain, or weaving irregularity. The output is a defect label, confidence or decision score where supported by the method, and the location of the suspected defect.

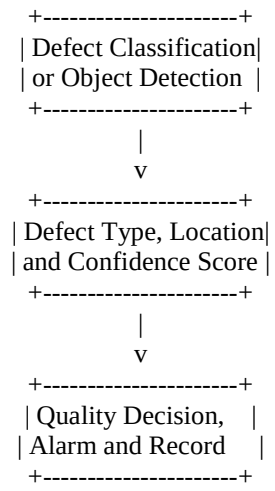
Stage 5: Result Recording and Notification

The input is the defect decision and associated location information. The system displays the inspection result, stores relevant information, and generates an alarm or marking signal if required. The output is an inspection record that can be used by quality-control personnel to locate defects, review recurring problems, and monitor fabric quality.

4. Pipeline 3: Deep-Learning-Based Computer Vision Pipeline

4.1 Block Diagram





4.2 Working Principle

A deep-learning-based pipeline uses a neural network to learn visual patterns from a dataset of fabric images. During development, the model is trained using suitable examples of defective and defect-free fabric. Depending on the chosen task, the training images may be labelled with defect categories, bounding boxes, or pixel-level masks. The model learns patterns associated with different defects, potentially reducing the need to design every feature manually. During actual inspection, the trained model receives a newly acquired fabric image and performs inference to predict whether defects are present and, depending on the model, identify their categories and locations.

A convolutional neural network (CNN) may be used for image classification, while an object-detection model may identify and locate individual defects. A segmentation model may be suitable when the exact shape or area of a defect must be identified. For example, a trained model may learn visual differences between normal weaving patterns and irregular regions associated with broken threads. It may also identify stains or holes when suitable training examples are available. However, deep learning requires representative training data, model development, evaluation, and suitable deployment resources. A model trained on one fabric type may perform less effectively on another fabric with different colours, textures, or weaving patterns. Furthermore, although deep learning can provide strong detection performance, its internal decision-making is generally less directly interpretable than explicit rule-based feature analysis.

4.3 Input and Output of Each Stage

Stage 1: Image Acquisition

The input is the moving fabric surface. The camera captures a digital image under suitable lighting conditions. The output is the raw image that will be evaluated by the trained neural network.

Stage 2: Image Preparation

The input is the raw image. The system applies the preprocessing operations required by the selected model, such as resizing, normalization, and conversion into the expected input format. Consistent preprocessing is important because the model must receive images in the same format used during training. The output is a model-ready image or tensor.

Stage 3: Neural-Network Inference

The input is the prepared image. The trained neural network processes the image through its learned layers and produces predictions. Depending on the model, these predictions may contain defect probabilities, class scores, bounding-box coordinates, or a segmentation map. The output is the model's raw prediction information.

Stage 4: Defect Classification and Localization

The input is the neural network's predictions. Post-processing operations may remove duplicate detections, apply confidence thresholds, and convert predicted regions into defect labels and coordinates. The output may identify a hole, broken thread, stain, or weaving irregularity, together with its location and confidence score. The quality of this output depends on the model, training dataset, image quality, and selected thresholds.

Stage 5: Quality Decision and Recording

The input is the processed defect prediction. The system determines whether the fabric should be accepted, flagged for review, marked, or rejected according to the established quality policy. It may also save selected defect images and inspection details. The output is a final inspection result and its associated record.

Important distinction: Training and inference are different activities. Training is performed during model development and can require substantial computing resources. Inference is the operation performed on new images during production. The 0.5-second decision constraint applies to the operational inspection pipeline, although image transfer and any required post-processing must also be included when measuring the actual response time.

5. Stages Required to Detect Different Fabric Defects

The four required defect types have different visual characteristics. Therefore, the inspection pipeline should use appropriate combinations of image enhancement, feature extraction, and decision-making rather than relying on one processing operation for every defect.

A. Hole Detection

A hole is a region where the expected fabric structure is missing or interrupted. The system should first acquire an image with sufficient spatial resolution and suitable illumination. Preprocessing and contrast enhancement may improve the visibility of the hole, while thresholding or segmentation may separate the suspected region from the surrounding fabric. Connected-component analysis can then measure the region's area, shape, and position. Edge or texture analysis may help distinguish a genuine hole from an ordinary dark pattern. A final decision rule or trained model classifies the candidate region as a hole. The system must be tested on different fabric colours and weave patterns because a dark region does not always indicate a physical hole.

B. Broken-Thread Detection

A broken thread may appear as a thin discontinuity in a yarn or as an interruption in the regular fabric pattern. The acquisition system must capture enough detail to represent the thread structure clearly. Noise reduction should be used cautiously because excessive smoothing may remove thin thread features. Edge detection, line analysis, local texture analysis, and comparison with the expected weave pattern can help identify the discontinuity. The resulting feature information is passed to the decision stage, which determines whether the irregularity is consistent with a broken thread. Deep-learning methods can also be trained to identify such defects if the training data contains sufficient examples of broken threads and normal fabric variations.

C. Oil-Stain Detection

Oil stains may create changes in brightness, colour, reflectance, or local texture. When a stain has low contrast against the surrounding material, the system may require controlled illumination and suitable contrast enhancement. If colour information helps distinguish the stain, it may be preferable to preserve the relevant colour channels rather than converting every image immediately to grayscale. Thresholding or colour-based analysis can identify candidate regions, while region statistics and texture features can help distinguish stains from normal variations in the fabric. The high-level decision stage classifies the

candidate as a stain or rejects it as a false alarm. Deep-learning models may also detect stains, but they should be trained and tested using representative examples of different stain sizes, colours, and lighting conditions.

D. Weaving-Irregularity Detection

Weaving irregularities involve deviations from the normal arrangement or repetition of threads. They may include uneven spacing, missing threads, abnormal patterns, or local changes in texture. Image acquisition must capture the regular structure of the fabric with sufficient detail. Texture analysis, local intensity statistics, frequency-based analysis, and comparisons with neighbouring fabric regions may help identify deviations from the expected pattern. Edge and line analysis may also reveal abnormal thread arrangements. The decision stage compares the extracted features with an expected pattern or learned reference and identifies regions that require inspection. Because normal fabric textures vary between materials, the system should be calibrated or configured for the fabric types being produced.

Summary of Required Processing Stages

Defect type	Important processing stages	Main features examined
Holes	Image acquisition, contrast enhancement, thresholding or segmentation, connected-component analysis, defect decision	Region area, shape, boundaries, missing fabric structure
Broken threads	Image acquisition, careful noise reduction, edge detection, line analysis, texture analysis, defect decision	Thin lines, discontinuities, broken yarn structure
Oil stains	Controlled illumination, colour or grayscale preprocessing as appropriate, contrast enhancement, thresholding or region analysis, classification	Intensity, colour, reflectance, local texture
Weaving irregularities	Image acquisition, preprocessing, texture analysis, edge or line analysis, comparison with expected pattern, defect decision	Repeated texture, thread spacing, pattern deviation

These stages describe suitable candidate techniques rather than a guarantee that each technique will detect every defect. The final combination should be chosen after testing on representative fabric samples.

6. Comparison of the Three Pipeline Designs

Comparison factor	Low-level pipeline	Three-level pipeline	Deep-learning-based pipeline

Main purpose	Improves image quality.	Extracts features and makes explicit defect decisions.	Learns patterns from data and predicts defects.
Processing stages	Acquisition, preprocessing, enhancement.	Acquisition, low-level processing, feature extraction, defect decision.	Acquisition, preparation, neural-network inference, classification or detection, post-processing.
Defect identification	Limited unless detection logic is added.	Supports defect identification through engineered features and rules.	Supports learned classification, detection, or segmentation, depending on the model.
Computational complexity	Generally low.	Low to moderate when efficient features and rules are used; may increase with complex texture analysis.	Often moderate to high, depending on model size, input resolution, and architecture.
Explainability	High for individual image transformations.	Generally high because features and decision rules can be inspected.	Usually lower because the learned internal representations are less directly interpretable.
Hardware requirements	Usually modest.	Can often operate on a standard CPU with an efficient implementation.	Depends on the model; lightweight inference may run on a CPU, while larger models may need acceleration.
Data requirements	Does not necessarily require labelled training data.	Rule-based methods may not require training data, but tuning and validation samples are needed.	Requires representative labelled data for supervised training, depending on the chosen learning method.
Main advantage	Simple, fast, and easy to implement.	Combines interpretable features with meaningful defect decisions.	Can learn complex visual patterns that are difficult to describe with fixed rules.
Main limitation	Does not independently classify defects without additional logic.	Hand-designed features may need adjustment for different fabrics and defects.	Needs suitable training data, model validation, and sufficient inference performance.

The comparison shows that the low-level pipeline has the lowest implementation complexity, but its output is primarily an enhanced image rather than a complete inspection decision. The deep-learning-based pipeline can learn complex patterns, but its computational requirements and data needs depend on the model selected. The three-level pipeline offers a practical balance by combining basic image

processing, explicit feature extraction, and interpretable defect decisions. However, its speed and accuracy must still be confirmed through testing, especially when processing fabric with complex patterns or low-contrast defects.

7. Recommended Pipeline for the Available Hardware

The recommended initial design is the three-level computer vision pipeline using optimized image processing, intermediate-level feature extraction, and rule-based or lightweight classification. This recommendation is based on the available four-core CPU, 16 GB RAM, and maximum decision time of 0.5 seconds. A low-level pipeline alone cannot reliably produce the required defect classifications, while a deep-learning model may introduce additional training and deployment requirements. A carefully designed three-level pipeline can use efficient operations such as grayscale conversion where appropriate, contrast enhancement, selective filtering, edge detection, connected-component analysis, and texture-based features to detect the specified defects. The high-level decision stage can then apply defect-specific rules or a lightweight classifier to distinguish likely holes, broken threads, stains, and weaving irregularities.

The proposed implementation should use Python, OpenCV, and NumPy to reduce the complexity of developing and testing the processing pipeline. Image processing should be optimized to avoid unnecessary operations, and only the information needed for defect analysis should be calculated at each stage. For example, a hole-detection routine may analyze connected regions, while a broken-thread routine may examine line discontinuities and local texture. Oil-stain detection may require contrast or colour analysis, whereas weaving-irregularity detection may depend more strongly on texture characteristics. Separating these tasks logically allows each defect type to use appropriate features without requiring every expensive operation to be applied to every image. The processing pipeline should also record defect coordinates and generate an inspection result so that the system can support quality-control decisions.

The 0.5-second constraint must be verified by measuring the complete operational response time, including image acquisition or transfer where applicable, preprocessing, feature extraction, decision-making, and any required post-processing. An algorithm that completes only its final classification within 0.5 seconds may still fail the requirement if the overall pipeline takes longer. Testing should be performed on the actual four-core CPU with representative image sizes and realistic fabric movement. The system should be evaluated for detection accuracy, missed defects, false alarms, memory usage, and processing

time. The stated 1.5 mm minimum target defect size and missed-defect target of no more than 2% must be treated as acceptance requirements that require experimental verification.

A lightweight deep-learning model may be considered as a later improvement if the initial feature-based pipeline cannot reliably detect complex weaving irregularities or defects that are difficult to describe using explicit rules. Before adoption, the model should be evaluated for CPU inference time, memory use, accuracy, and generalization across different fabric types. If it meets the performance requirement on the available hardware, it could be integrated into the three-level system as an additional classifier or compared with the rule-based method. This hybrid approach should be adopted only if experiments show that it improves detection sufficiently without violating the decision-time constraint.

TASK 3: IMAGE REPRESENTATION ANALYSIS

1. Introduction

Image representation is the method used to store and describe the visual information contained in a digital image. In computer vision-based textile inspection, the choice of image representation affects how effectively the system identifies defects such as colour variations, oil stains, holes, broken threads, and weaving irregularities. A digital image consists of pixels, and each pixel stores numerical information about the appearance of a small region of the fabric. Different representations store different types of visual information and require different amounts of memory and processing. The three commonly used representations considered in this task are RGB colour, HSV colour, and grayscale. RGB represents colours using red, green, and blue components, HSV represents colour using hue, saturation, and value,

and grayscale represents image intensity using a single channel. Selecting an appropriate representation for Sri Lakshmi Textile Industries requires balancing defect visibility, processing complexity, memory usage, and the need to distinguish colour-based defects from structural defects.

2. RGB Colour Representation

2.1 How RGB Stores Image Information

RGB stands for **Red, Green, and Blue**, the three colour components used to represent a wide range of colours in a digital image. In a typical 8-bit RGB image, each pixel stores three numerical values: one for red, one for green, and one for blue. Each component generally ranges from 0 to 255, where 0 represents the minimum intensity of that component and 255 represents the maximum intensity. Different combinations of these values produce different colours. For example, a pixel with values of (255, 0, 0) represents pure red, while (0, 255, 0) represents pure green and (0, 0, 255) represents pure blue. A pixel with equal component values, such as (120, 120, 120), represents a shade of grey. An RGB image therefore preserves colour information that may be useful for identifying changes in fabric colour, stains, dyeing inconsistencies, and other colour-related defects.

In the textile inspection system, the camera can capture an RGB image of the fabric and provide it to the image processing pipeline. The three colour channels allow the system to compare the colour of a suspicious region with the surrounding material. For example, a dark oil stain on a light-coloured fabric may produce a noticeable difference in its RGB values. Similarly, an unwanted colour patch or uneven dyeing region may be detected by comparing the colour components across neighbouring pixels. However, RGB values are also affected by illumination, shadows, camera exposure, and the reflectance properties of the material. A change in lighting can alter the RGB values even when the actual fabric colour remains unchanged. Therefore, RGB-based defect detection may require suitable lighting, calibration, and preprocessing to reduce unwanted variations.

2.2 Suitability of RGB for Fabric Defect Detection

RGB representation is particularly useful for detecting **colour variations and stains** because it retains the colour information that grayscale images discard. If a fabric contains an unwanted dye mark, coloured contamination, or a stain that differs from the background in its colour components, RGB analysis may help distinguish the affected region. RGB information may also assist when different types of fabric defects have different colours. However, RGB is not always the simplest representation for identifying structural defects. Holes and broken threads may be more easily detected through their shape, edges,

intensity differences, or texture patterns. Although RGB images can also be used for these tasks, processing all three channels may increase computational demand without always providing additional useful information. Consequently, RGB is valuable when colour is an important indicator of fabric quality, but it may not be necessary for every stage of the inspection pipeline.

3. HSV Colour Representation

3.1 How HSV Stores Image Information

HSV stands for **Hue, Saturation, and Value**. Unlike RGB, which represents a colour through the intensity of three primary colour components, HSV separates colour information into three conceptually different components. Hue represents the basic colour, such as red, green, yellow, or blue. Saturation represents the strength or purity of the colour, with lower saturation generally indicating a more washed-out or greyish appearance. Value represents the brightness of the pixel. This separation can make certain colour analysis tasks easier because the system can examine colour characteristics separately from brightness. In common software implementations, HSV values are stored numerically, although their ranges depend on the data type and implementation. For example, some OpenCV operations use a reduced hue range for 8-bit images, while other formats use a wider numerical range.

For textile inspection, HSV can help identify colour differences that may be difficult to isolate by examining RGB components independently. A coloured stain may produce a different hue or saturation from the surrounding fabric, while a darker stain may produce a lower value. The system can analyze these characteristics to identify candidate regions that differ from the expected fabric appearance. HSV can also be useful when fabric brightness changes moderately but the stain retains a distinguishable hue or saturation. Nevertheless, HSV does not completely remove the effects of illumination. Changes in lighting can still affect saturation and value, and hue can become unstable in regions with very low saturation. Therefore, HSV should be used with suitable illumination and tested against representative fabric samples rather than assumed to be completely independent of lighting conditions.

3.2 Suitability of HSV for Fabric Defect Detection

HSV representation is particularly suitable for detecting **colour variations and certain oil stains** when the defect differs from the surrounding fabric in hue or saturation. For example, an unwanted coloured mark on a white or light-coloured fabric may be identified by examining changes in saturation, while a coloured stain may be separated from the normal fabric using a suitable hue range. The value component can also help locate unusually dark regions. However, not every oil stain produces a clear hue difference.

Some oil stains mainly alter brightness or surface reflectance, so they may require value-based analysis, texture analysis, or carefully controlled lighting. Holes and broken threads are generally structural defects, meaning their important characteristics are related to missing material, edges, line discontinuities, or changes in weaving texture. HSV can be used to process their images, but it may not provide a major advantage over grayscale for these defects. Its main strength is the separation of colour-related information into components that can be analyzed individually.

4. Grayscale Representation

4.1 How Grayscale Stores Image Information

A grayscale image represents each pixel using a single intensity value rather than separate red, green, and blue components. In a typical 8-bit grayscale image, each pixel has a value between 0 and 255. A value of 0 represents black, a value of 255 represents white, and intermediate values represent different shades of grey. When an RGB image is converted to grayscale, the colour components are combined into a single intensity value using a conversion formula. A common approximate luminance-based conversion is:

Grayscale = 0.299R + 0.587G + 0.114B

Here, R, G, and B represent the red, green, and blue components of a pixel. The formula gives different weights to the three channels because they contribute differently to perceived brightness. The conversion reduces the number of image channels from three to one, which lowers memory usage and can simplify later processing operations. However, the conversion also removes explicit colour information. Two differently coloured pixels can produce similar grayscale intensities, so a colour defect may become less distinguishable after conversion.

4.2 Suitability of Grayscale for Fabric Defect Detection

Grayscale representation is particularly useful for detecting **holes, broken threads, and weaving irregularities** when these defects produce changes in brightness, shape, edges, or texture. For example, a hole may appear as an unusually dark or bright region depending on the background behind the fabric and the illumination arrangement. A broken thread may create a discontinuity in the normal fabric pattern, while a weaving irregularity may produce a local change in texture. Grayscale images can be processed using thresholding, edge detection, filtering, connected-component analysis, and texture analysis. These operations can often be performed efficiently because the system processes one intensity channel instead of three colour channels. Grayscale may also help identify oil stains when they create a noticeable

brightness difference from the surrounding fabric. However, it may fail to distinguish a coloured stain from normal fabric if both have similar grayscale intensities despite having different colours. Therefore, grayscale is efficient for structural defect detection but should not automatically replace colour representations when colour itself is important for identifying defects.

5. Comparison of RGB, HSV, and Grayscale Representations

The three image representations preserve different information and therefore have different strengths in textile defect inspection. RGB preserves the original colour components, HSV separates colour characteristics into hue, saturation, and value, and grayscale reduces the image to a single intensity channel. Their suitability depends on the defect type and the amount of visual information needed for reliable detection.

Comparison factor	RGB representation	HSV representation	Grayscale representation
Information stored	Red, green, and blue colour components.	Hue, saturation, and value components.	One intensity value per pixel.
Number of channels	Three.	Three.	One.
Colour information	Preserves colour directly.	Represents colour through hue and saturation, with brightness represented separately by value.	Explicit colour information is lost.
Colour variation detection	Very suitable because all three colour components are retained.	Very suitable when colour differences can be identified through hue or saturation.	Limited when different colours have similar brightness.
Oil-stain detection	Useful when the stain produces colour or intensity differences.	Useful when the stain produces hue, saturation, or value differences.	Useful when the stain produces a clear intensity difference.
Hole detection	Possible, but colour information may not always be necessary.	Possible, but colour components may not provide a major advantage.	Often suitable when the hole produces a clear intensity, shape, or edge difference.

Comparison factor	RGB representation	HSV representation	Grayscale representation
Broken-thread detection	Possible using edges and texture, but all channels may not be needed.	Possible, although hue and saturation may provide limited benefit for structural defects.	Suitable for edge, line-discontinuity, and texture analysis.
Weaving-irregularity detection	Useful when colour and texture both contribute to the defect.	Useful when colour differences need to be separated from brightness information.	Suitable for many texture and structural analyses.
Memory requirement	Relatively high for the same image dimensions and bit depth.	Similar to RGB when all three channels use the same storage depth.	Lower because only one channel is stored.
Processing complexity	Moderate because three channels may need to be processed.	Moderate because colour conversion and component-specific analysis may be required.	Usually lower for basic processing because only one channel is handled.
Main advantage	Retains detailed colour information.	Separates colour characteristics from brightness conceptually.	Efficient for many structural and intensity-based operations.
Main limitation	Sensitive to illumination changes and may require processing across several channels.	Still affected by lighting, and hue can be unreliable for weakly saturated pixels.	Cannot directly distinguish colours that become similar in intensity.

6. Memory Requirements and Processing Complexity

For the same image dimensions and an 8-bit storage format, an RGB image requires approximately three bytes per pixel because it stores one byte for each of the red, green, and blue components. A standard 8-bit grayscale image requires approximately one byte per pixel because it stores a single intensity value. An HSV image using three 8-bit channels also requires approximately three bytes per pixel. Therefore, converting an RGB image to HSV does not automatically reduce memory usage; both normally contain three channels. The principal difference between RGB and HSV is how the colour information is represented and interpreted, not the number of stored channels. Actual memory use may be higher because software can create temporary arrays, intermediate images, or higher-precision representations during processing.

For example, consider a fabric image with dimensions of 1200 × 400 pixels. It contains 480,000 pixels. If each channel uses 8 bits, the approximate raw pixel storage requirements are as follows:

- **RGB:** 480,000 × 3 bytes = 1,440,000 bytes, approximately 1.44 MB.
- **HSV:** 480,000 × 3 bytes = 1,440,000 bytes, approximately 1.44 MB.
- **Grayscale:** 480,000 × 1 byte = 480,000 bytes, approximately 0.48 MB.

These estimates describe uncompressed pixel storage and exclude file headers, alignment, temporary processing arrays, and other software overhead. If the specified 1200 × 400 mm field of view is interpreted as a physical area rather than a pixel resolution, the actual image dimensions must be determined from the camera's resolution and calibration before calculating the storage requirements. The figures above assume a 1200 × 400-pixel image solely to illustrate the memory difference.

In terms of processing complexity, grayscale generally requires less memory bandwidth and fewer operations for basic intensity-based processing because only one channel is handled. Operations such as thresholding, edge detection, and connected-component analysis can therefore be relatively efficient on grayscale images. RGB processing may require calculations across three channels, increasing the computational load when every channel is processed independently. HSV may require a colour-space conversion from RGB before hue, saturation, or value can be analyzed, adding an additional operation. However, the actual speed depends on the algorithm, image size, software implementation, and hardware. Processing a single relevant HSV channel may be more efficient than applying a complex operation to all RGB channels, so it is important to compare complete processing pipelines rather than assume that one representation is always faster.

7. Recommended Image Representation for Sri Lakshmi Textile Industries

The recommended approach is a selective combination of grayscale and HSV, with RGB retained when detailed colour information is required. This combination provides a practical balance between computational efficiency and the ability to detect both structural and colour-related defects. Grayscale should be used as the primary representation for many structural inspection tasks, including hole detection, broken-thread detection, edge analysis, and weaving-irregularity analysis. It reduces the image to one intensity channel and supports efficient operations such as filtering, thresholding, edge detection, and texture analysis. This is useful for the available four-core CPU system because reducing unnecessary processing can help the pipeline meet the required decision time of 0.5 seconds.

HSV should be used when the system needs to identify defects whose colour characteristics differ from the surrounding fabric. For example, if an unwanted dye mark has a distinguishable hue or saturation, the system can analyze the relevant HSV components to locate the suspicious region. For low-contrast oil stains, the value component may provide useful brightness information, although additional contrast enhancement or controlled lighting may be needed. RGB can be retained for cases in which the original colour components provide useful information that is not adequately captured by the selected HSV or grayscale features. The system should avoid converting every image into every representation without a clear purpose, because unnecessary conversions and duplicate image storage can increase processing time and memory usage.

The proposed processing strategy is therefore to acquire a colour image when colour information is needed, generate a grayscale representation for structural analysis, and use HSV selectively for colour-based defect detection. The results from these processing paths can then be combined at the decision stage. For instance, grayscale analysis may identify a suspicious region with an unusual shape or texture, while HSV analysis may determine whether that region also has an unusual colour or saturation. Combining these results can help distinguish different defect types more effectively than relying on one representation alone. However, the effectiveness of this strategy must be verified using representative fabric samples, particularly when oil stains have very low contrast or when different fabric colours produce different intensity patterns.

For the initial production conditions, the system must detect defects as small as 1.5 mm and produce an inspection decision within 0.5 seconds. The selected representation should therefore be evaluated not only for visual suitability but also for detection accuracy, missed-defect rate, false alarms, memory consumption, and processing time. The system should use the simplest representation that preserves the information required for each defect type. If grayscale performs adequately for holes and broken threads, those operations need not process all three colour channels. If HSV provides a measurable improvement in detecting colour variations or stains, it can be used for those tasks specifically. This selective approach avoids unnecessary computation while retaining the information needed for accurate inspection.

Conclusion

RGB, HSV, and grayscale are three useful image representations for textile fabric inspection, but each serves a different purpose. RGB preserves the original colour components and is valuable for detecting colour variations. HSV separates hue, saturation, and brightness-related information, making it useful for selected colour-based inspection tasks. Grayscale uses a single intensity channel and is generally efficient

for identifying structural defects such as holes, broken threads, and weaving irregularities. RGB and HSV normally require similar storage when represented with three 8-bit channels, whereas grayscale requires approximately one-third as much raw pixel storage. For Sri Lakshmi Textile Industries, a selective combination of grayscale for structural analysis and HSV for colour-related analysis is recommended, with RGB retained where it provides additional useful information. This approach balances defect visibility, memory requirements, and computational complexity while supporting the 0.5-second decision target, subject to practical testing and validation.

TASK 4: CONNECTIVITY ANALYSIS

1. Introduction

Connectivity analysis is an important concept in digital image processing used to determine whether pixels belong to the same connected region. In computer vision-based textile inspection, connectivity analysis helps identify and measure defect regions such as small holes, broken threads, oil stains, and irregular weaving patterns. After a fabric image is converted into a binary image through thresholding or segmentation, defect pixels may be represented by 1 and background pixels by 0. Connectivity rules determine whether neighbouring defect pixels should be grouped into a single connected component or treated as separate regions. The three common connectivity approaches are 4-connectivity, 8-connectivity, and mixed connectivity (m-connectivity). Each method uses a different definition of pixel neighbourhood, which affects the number of connected components, their boundaries, and their measured sizes. Selecting an appropriate connectivity method is important for Sri Lakshmi Textile Industries because the system must detect small defects accurately while avoiding incorrect merging or splitting of defect regions.

2. Definition of Pixel-Connectivity Approaches

2.1 Four-Connectivity (4-Connectivity)

Four-connectivity considers only the four pixels that share an edge with the central pixel. These neighbours are located directly above, below, to the left, and to the right. Diagonal pixels are not considered connected under this method. If two foreground pixels share an edge, they may belong to the same connected component, provided that a chain of such edge-connected foreground pixels links them.

Neighbour diagram for 4-connectivity:

```
+---+---+---+
| 0 | 1 | 0 |
+---+---+---+
| 1 | P | 1 |
+---+---+---+
| 0 | 1 | 0 |
+---+---+---+
```

Legend:

- P = central pixel.
- 1 = neighbour considered under 4-connectivity.
- 0 = diagonal neighbour not considered.

The 4-connected neighbourhood of a pixel at position (x,y)(x,y) can be represented as:

$$N4(x,y)=\{(x-1,y),(x+1,y),(x,y-1),(x,y+1)\}$$

Four-connectivity is useful when only direct horizontal and vertical connections should join pixels. In fabric inspection, it may prevent two regions that touch only at their corners from being merged. However, it may divide a genuine irregular defect into multiple components when its pixels are connected only diagonally.

2.2 Eight-Connectivity (8-Connectivity)

Eight-connectivity considers all eight surrounding pixels, including the four edge-sharing neighbours and the four diagonal neighbours. Therefore, two foreground pixels may belong to the same component if they share an edge or a corner, di

TASK 5: IMAGE ACQUISITION ANALYSIS

Introduction

Image acquisition is the process of capturing images of an object using a camera and converting the captured light into digital image data for further processing. In textile industries, image acquisition is an important stage of automated fabric inspection because the quality of the captured image directly affects the ability to detect defects such as holes, broken threads, oil stains, and weaving irregularities.

In Sri Lakshmi Textile Industries, the fabric moves continuously along the production line at an initial speed of **0.6 m/s**. The fabric width is approximately **1000 mm**, and the camera has a field of view of **1200 × 400 mm**. The inspection system must detect defects as small as **1.5 mm**, produce a decision within **0.5 seconds**, and support continuous fabric inspection. Therefore, the camera must capture clear images without introducing significant motion distortion or losing small defect details.

Three camera acquisition techniques are considered: rolling-shutter cameras, global-shutter cameras, and line-scan cameras.

1. Image Acquisition Principle of Each Camera Type

1.1 Rolling-Shutter Camera

A rolling-shutter camera captures an image by exposing different rows of pixels at slightly different times. Instead of capturing the entire image simultaneously, the sensor scans the image row by row. The first row is exposed before the next row, and this process continues until the complete image is captured.

Working principle:

1. Light reflected from the moving fabric enters the camera lens.
2. The image sensor begins capturing the first row of pixels.
3. Subsequent rows are captured at slightly different times.
4. The sensor combines the captured rows to form a complete image.
5. The image is transferred to the computer for defect detection.

Advantages:

- Generally available at a relatively low cost.

- Suitable for many stationary or slowly moving objects.
- Can provide good image quality under suitable lighting conditions.
- Widely used in general-purpose industrial imaging applications.

Limitations:

- Different rows represent different moments in time.
- Moving fabric can appear stretched, compressed, tilted, or distorted.
- Fast-moving defects may not retain their original shape.
- Distortion can reduce the accuracy of defect measurements.

Application to the textile line: At 0.6 m/s, the fabric travels while the camera scans successive rows. Consequently, different parts of the captured image may correspond to different fabric positions. The severity of distortion depends on the sensor's readout time, exposure time, and the direction of movement relative to the sensor.

1.2 Global-Shutter Camera

A global-shutter camera exposes all pixels in the image sensor at approximately the same time. The entire image represents a common instant, even though the sensor may read out the captured data afterward.

Working principle:

1. Light reflected from the fabric enters the camera lens.
2. All sensor pixels begin exposure together.
3. The camera captures the fabric's appearance at approximately the same instant.
4. The captured image is read out and transferred to the computer.
5. Image-processing algorithms analyse the image to identify defects.

Advantages:

- Reduces geometric distortion caused by differences in row-exposure timing.
- Preserves the shape and position of moving defects more accurately than a rolling-shutter camera under comparable conditions.
- Suitable for high-speed industrial inspection.
- Supports accurate measurement when paired with suitable optics, lighting, and calibration.

Limitations:

- May cost more than a comparable general-purpose rolling-shutter camera.
- Requires an appropriate exposure time to prevent motion blur.
- A single camera may not provide sufficient spatial resolution across the entire fabric width.
- A frame-based system may require careful control of frame rate, lighting, and image-processing time.

Application to the textile line: A global-shutter camera is a suitable option when the system needs to capture the entire fabric region at once. However, the exposure time must be sufficiently short to prevent the moving fabric from travelling too far during image capture.

1.3 Line-Scan Camera

A line-scan camera captures one line of pixels at a time instead of capturing a complete two-dimensional image in a single exposure. As the fabric moves beneath the camera, the system captures successive lines and combines them to construct a two-dimensional image of the fabric.

Working principle:

1. A line-scan camera is mounted above the moving fabric.
2. The camera captures a single line across the fabric width.
3. As the fabric advances, the camera captures another line at a controlled interval.
4. Successive lines are assembled in sequence.
5. The resulting image represents the fabric surface along its direction of travel.
6. The inspection software analyses the reconstructed image for defects.

Advantages:

- Well suited to continuous materials such as fabric, paper, plastic film, and metal sheets.
- Can inspect long lengths of fabric without requiring a conventional full-frame image of the entire inspection area.
- Provides high spatial resolution across the width when the sensor resolution and optics are selected appropriately.
- Can support detailed inspection of small defects over a large fabric surface.

- Integrates naturally with production-line movement when line acquisition is synchronised with fabric travel.

Limitations:

- Requires careful synchronisation with fabric movement, often using an encoder.
- Installation and calibration can be more complex than for a basic area-scan camera.
- Fabric speed variations can cause stretching or compression along the travel direction if line timing is not adjusted correctly.
- The camera, lighting, frame grabber or interface, and processing hardware may increase system cost.

Application to the textile line: Because fabric moves continuously through the inspection area, a line-scan camera can capture consecutive lines across the full width and build a continuous surface image. If the line rate is synchronised with the fabric speed, it can preserve defect shape and support reliable inspection of small defects.

2. Effect of Fabric Movement at 0.6 m/s on Image Distortion and Defect Visibility

The fabric speed is **0.6 m/s**, which is equivalent to **600 mm/s**. Therefore, the fabric moves 600 mm in one second.

The movement during an exposure is calculated as:

Movement during exposure = Fabric speed × Exposure time

For example, if the exposure time is 1 millisecond:

Movement = 600 mm/s × 0.001 s

Movement = **0.6 mm**

This means that the fabric moves approximately 0.6 mm while the image is being exposed. Since the smallest required defect is 1.5 mm, this movement is significant enough to require careful exposure-time control.

If the exposure time is reduced to 0.2 milliseconds:

Movement = 600 mm/s × 0.0002 s

Movement = **0.12 mm**

This shorter exposure substantially reduces motion blur, although sufficient illumination is needed to maintain image quality.

2.1 Rolling-Shutter Camera and Fabric Movement

In a rolling-shutter camera, different rows are captured at different times. Since the fabric moves continuously, its position changes while the camera reads the image. This can produce geometric distortion in addition to motion blur during each row's exposure.

Possible effects include:

- A circular hole may appear stretched or tilted.
- Broken threads may appear displaced or distorted.
- The apparent width of a defect may change.
- Small defects may become more difficult to measure accurately.
- Low-contrast oil stains may become less distinguishable if movement and exposure conditions reduce image clarity.

The severity of these effects depends on the sensor's row-readout timing, exposure duration, and fabric movement. A rolling-shutter camera is not automatically unsuitable, but its performance must be validated under actual operating conditions.

2.2 Global-Shutter Camera and Fabric Movement

A global-shutter camera exposes all pixels at approximately the same time, reducing geometric distortion associated with sequential row exposure.

However, the fabric continues to move during the exposure. If the exposure is too long, a defect can still produce motion blur.

For example, at 0.6 m/s:

- A 1 ms exposure produces approximately 0.6 mm of movement.
- A 0.2 ms exposure produces approximately 0.12 mm of movement.

A short exposure helps preserve defect boundaries. Adequate lighting and sufficient image resolution are also necessary.

2.3 Line-Scan Camera and Fabric Movement

A line-scan camera relies on successive line captures as the fabric moves. The line rate determines how frequently the system samples the fabric along its direction of travel.

At 0.6 m/s, the fabric advances 0.6 mm in every 1 millisecond. If the camera captures 1,000 lines per second, the fabric advances 0.6 mm between successive lines. This sampling interval may be too large for reliable inspection of a 1.5 mm defect, depending on the required number of samples across the defect and the image-processing method.

For example, if the design target is to obtain at least five samples across a 1.5 mm defect along the direction of travel, the sampling interval should be no greater than:

Sampling interval = 1.5 mm ÷ 5

Sampling interval = **0.3 mm**

The corresponding minimum line rate at 0.6 m/s is:

Line rate = 600 mm/s ÷ 0.3 mm

Line rate = **2,000 lines per second**

Thus, a line rate of at least 2,000 lines per second would provide five line samples across a 1.5 mm defect along the direction of travel under these assumptions. In practice, the selected rate should include an appropriate margin and must also satisfy the sensor's exposure, illumination, and data-transfer requirements.

2.4 Comparison of Motion Effects

Factor	Rolling-Shutter Camera	Global-Shutter Camera	Line-Scan Camera
Image capture	Rows captured at different times	All pixels exposed at approximately the same time	One line captured at a time

Factor	Rolling-Shutter Camera	Global-Shutter Camera	Line-Scan Camera
Effect of fabric movement	May cause geometric distortion and blur	Primarily affected by motion during exposure	Depends on line rate and synchronisation
Defect shape	May be distorted during row acquisition	Generally preserves shape more accurately	Can preserve shape when line timing is correctly controlled
Small-defect visibility	Depends on exposure, readout time, and resolution	Depends on exposure, resolution, and lighting	Depends on across-width resolution and line sampling
Continuous inspection	Possible, but requires validation	Possible, but frame coverage must be adequate	Particularly well suited to continuous web inspection

3. Comparison of Image Quality, Suitability, Installation Complexity, and Cost

Comparison Factor	Rolling-Shutter Camera	Global-Shutter Camera	Line-Scan Camera
Image quality on moving fabric	Can be good, but rolling-shutter distortion may occur	Good temporal consistency across the image; motion blur remains possible	High-quality continuous images are possible with suitable optics, illumination, and line timing
Continuous web inspection	Possible, depending on frame rate and coverage	Possible, depending on frame rate and field of view	Very suitable for continuous fabric, paper, and film inspection
Detection of 1.5 mm defects	Possible if resolution and exposure are sufficient, but distortion must be assessed	Suitable if spatial resolution and exposure are sufficient	Suitable if line rate, spatial resolution, and synchronisation are sufficient
Installation complexity	Generally low to moderate	Moderate	Moderate to high
Synchronisation requirements	Exposure and frame timing must be managed	Exposure and frame timing must be managed	Line acquisition must be synchronised with fabric motion
Hardware cost	Generally lower	Generally moderate to high	Often higher, depending on sensor resolution and interface
Processing requirements	Depends on frame resolution and frame rate	Depends on frame resolution and frame rate	Depends on line rate, image width, and image reconstruction
Main limitation	Sequential row capture can distort moving objects	Motion blur can still occur during exposure	Incorrect line timing can stretch or compress the reconstructed image

Note: The cost descriptions are relative and indicative. Actual costs depend on the sensor, resolution, optics, interface, lighting, software, and installation requirements. A quotation and performance test are needed before finalising the budget.

4. Selection of a Suitable Camera for Sri Lakshmi Textile Industries

Recommended camera type: Line-scan camera.

A line-scan camera is the preferred option for the proposed textile inspection system because the fabric moves continuously through the production line. It captures successive lines across the fabric width and reconstructs them into a continuous two-dimensional image. This method is well suited to inspecting long fabric surfaces for holes, broken threads, stains, and weaving irregularities.

The recommendation is based on the following factors:

4.1 Continuous Fabric Inspection

The fabric width is approximately 1000 mm, while the camera field of view is 1200 × 400 mm. A suitable line-scan system can be selected to cover the required width, with appropriate allowance for fabric-position variation and calibration.

Unlike a conventional area-scan system, a line-scan camera does not need to capture the entire inspection region as a complete two-dimensional frame at each instant. It builds the image line by line as the fabric advances. This makes it particularly suitable for continuous web inspection.

4.2 Detection of Small Defects

The system must detect defects as small as 1.5 mm. To achieve this, the camera must provide adequate spatial resolution across the fabric width and sufficient sampling along the direction of movement.

For example, suppose the camera is configured to capture 1200 pixels across a 1200 mm field of view. The nominal spatial sampling would be:

Spatial sampling = 1200 mm ÷ 1200 pixels

= **1 mm per pixel**

At this illustrative sampling level, a 1.5 mm defect would occupy approximately 1.5 pixels across the relevant dimension. This is generally too limited for robust detection and measurement because the defect may have weak contrast, fall between pixels, or be affected by optical blur and noise.

Therefore, the system should use a higher effective spatial resolution. For example, a target sampling interval of 0.3 mm per pixel would represent a 1.5 mm defect with approximately five pixels across that dimension.

For a 1200 mm field of view, this would require approximately:

1200 mm ÷ 0.3 mm/pixel = **4000 pixels across the width**

This is an illustrative design target, not a final camera specification. The actual requirement must be established through optical calibration and testing using representative 1.5 mm defects, including low-contrast stains.

4.3 Requirement for a 0.5-Second Decision

The inspection system must make a defect decision within 0.5 seconds. A line-scan camera can support this requirement when it is combined with suitable image acquisition, data transfer, image processing, and decision logic.

However, the camera type alone does not guarantee that the decision deadline will be met. The implementation should:

- Select a line rate suitable for the fabric speed and target defect size.
- Use adequate lighting to capture clear defect details.
- Process incoming image data efficiently.
- Avoid unnecessary storage and processing delays in the real-time inspection path.
- Measure end-to-end latency from image acquisition to defect decision.
- Confirm performance using representative fabric samples and the intended hardware.

The existing four-core CPU and 16 GB RAM may be sufficient for an appropriately designed classical image-processing pipeline, but this must be verified through benchmarking at the selected camera resolution and line rate.

4.4 Synchronisation with Fabric Movement

An encoder or another reliable measurement of fabric movement should be used to synchronise line acquisition with the production line. If fabric speed changes, the acquisition rate should be adjusted accordingly. Otherwise, the reconstructed image may appear stretched or compressed along the direction of travel.

The camera installation should also include stable mounting, suitable optics, uniform illumination, and calibration procedures. These measures help ensure consistent defect visibility throughout the fabric width.

4.5 Cost and Implementation Considerations

A line-scan camera may have higher initial equipment and integration costs than a basic rolling-shutter camera. However, the selection should be based on the overall inspection requirement rather than camera price alone.

The complete system cost must include the camera, lens, lighting, mounting, synchronisation hardware, data interface, processing computer, and software. The proposed system should be evaluated against the stated ₹150,000 budget. If the required resolution and acquisition rate cannot be achieved within that budget, the design should be revised using verified component specifications rather than assuming the budget is sufficient.

5. Justification Based on the 1.5 mm Defect Size and 0.5-Second Decision Requirement

The line-scan camera is recommended because it is naturally suited to continuous moving fabric and can provide dense spatial sampling over long lengths of material. Its suitability for the stated requirements depends on the following engineering conditions:

1. **Minimum defect size:** The camera and lens must provide enough effective pixels across a 1.5 mm defect. A practical initial target is approximately five pixels across the smallest defect dimension, followed by validation using real fabric samples. Low-contrast oil stains may require stronger illumination, suitable spectral lighting, and carefully selected image-processing methods.
2. **Fabric speed:** At 0.6 m/s, the line rate must be high enough to sample the fabric adequately along the direction of movement. A line rate of 2,000 lines per second gives a 0.3 mm travel distance between lines, corresponding to five line samples across a 1.5 mm defect under the stated assumptions.

3. **Image clarity:** Short exposure times and suitable lighting are required to reduce motion blur. The exposure time must be selected according to the acceptable movement during exposure and the sensor's light sensitivity.
4. **Decision deadline:** The complete system must acquire the relevant image data, process it, classify the defect, and generate a decision within 0.5 seconds. This must be confirmed through measured end-to-end latency testing.
5. **Production reliability:** Encoder-based synchronisation, stable mounting, camera calibration, and consistent illumination are needed to prevent image distortion and maintain repeatable detection performance.
6. **Budget feasibility:** The camera and supporting hardware must be selected only after confirming that the necessary spatial resolution, line rate, interface bandwidth, and illumination can be achieved within the available budget.

Conclusion

Rolling-shutter, global-shutter, and line-scan cameras use different image acquisition methods, and each has advantages and limitations when inspecting moving fabric. Rolling-shutter cameras can be economical but may introduce geometric distortion because different rows are captured at different times. Global-shutter cameras capture all pixels at approximately the same time, reducing this source of distortion, although motion blur can still occur. Line-scan cameras capture one line at a time and reconstruct a continuous image as the fabric moves, making them particularly suitable for textile web inspection.

For Sri Lakshmi Textile Industries, a **line-scan camera is the recommended initial design choice** because the fabric moves continuously and the system must detect defects as small as 1.5 mm. The selection should be finalised only after verifying the camera's spatial resolution, line rate, exposure requirements, illumination, data-transfer capacity, total system cost, and end-to-end decision time. A validated configuration can then be tested against the required 0.5-second decision deadline and the target missed-defect rate.

TASK 6: SAMPLING ANALYSIS

Marks: 5

Introduction

Image sampling is the process of converting a continuous visual scene into a digital image made up of individual pixels. The sampling resolution determines how much spatial detail can be represented in the captured image. Higher resolution generally provides more pixels to represent small defects, but it also increases storage requirements, data-transfer requirements, and processing time.

In Sri Lakshmi Textile Industries, the computer vision system must inspect fabric approximately 1000 mm wide using a camera field of view of 1200 mm across the width. The minimum defect size to be detected is 1.5 mm, and the system must produce a decision within 0.5 seconds. Three image resolutions are evaluated: 1280 × 1024 pixels, 2448 × 2048 pixels, and 4096 × 3000 pixels.

The aim is to select a resolution that provides adequate defect visibility while remaining practical for the available computing resources.

1. Calculation of Total Pixels for Each Resolution

The total number of pixels in a digital image is calculated using:

Total pixels = Image width in pixels × Image height in pixels

1.1 Low Resolution: 1280 × 1024 Pixels

Given:

- Image width = 1280 pixels
- Image height = 1024 pixels

Calculation:

Total pixels = 1280 × 1024

Total pixels = **1,310,720 pixels**

Therefore, the low-resolution image contains approximately **1.31 megapixels (MP)**.

1.2 Medium Resolution: 2448 × 2048 Pixels

Given:

- Image width = 2448 pixels
- Image height = 2048 pixels

Calculation:

Total pixels = 2448 × 2048

Total pixels = **5,013,504 pixels**

Therefore, the medium-resolution image contains approximately **5.01 megapixels (MP)**.

1.3 High Resolution: 4096 × 3000 Pixels

Given:

- Image width = 4096 pixels
- Image height = 3000 pixels

Calculation:

Total pixels = 4096 × 3000

Total pixels = **12,288,000 pixels**

Therefore, the high-resolution image contains approximately **12.29 megapixels (MP)**.

1.4 Pixel Calculation Summary

Resolution category	Image dimensions	Total pixels	Approximate megapixels
Low	1280 × 1024	1,310,720	1.31 MP
Medium	2448 × 2048	5,013,504	5.01 MP
High	4096 × 3000	12,288,000	12.29 MP

The high-resolution image contains approximately 9.38 times as many pixels as the low-resolution image. Consequently, it requires more memory and processing than the low-resolution option.

2. Calculation of Spatial Sampling Interval Across the 1200 mm Field of View

The spatial sampling interval indicates the physical distance represented by one pixel. For this assignment, it is calculated across the horizontal image width, assuming that the image width covers the entire 1200 mm field of view.

The formula is:

Spatial sampling interval = Physical field-of-view width ÷ Image width in pixels

Given:

Physical field-of-view width = 1200 mm

2.1 Low Resolution: 1280 × 1024 Pixels

Image width = 1280 pixels

Spatial sampling interval = 1200 mm ÷ 1280 pixels

= **0.9375 mm/pixel**

Therefore, one pixel represents approximately 0.938 mm across the fabric width.

2.2 Medium Resolution: 2448 × 2048 Pixels

Image width = 2448 pixels

Spatial sampling interval = 1200 mm ÷ 2448 pixels

= **0.4902 mm/pixel** (approximately)

Therefore, one pixel represents approximately 0.490 mm across the fabric width.

2.3 High Resolution: 4096 × 3000 Pixels

Image width = 4096 pixels

Spatial sampling interval = 1200 mm ÷ 4096 pixels

= **0.2930 mm/pixel** (approximately)

Therefore, one pixel represents approximately 0.293 mm across the fabric width.

2.4 Spatial Sampling Summary

Resolution	Horizontal image width	Physical field of view	Spatial sampling interval
Low	1280 pixels	1200 mm	0.9375 mm/pixel
Medium	2448 pixels	1200 mm	0.4902 mm/pixel
High	4096 pixels	1200 mm	0.2930 mm/pixel

The high-resolution image has the smallest sampling interval. Therefore, it can represent smaller spatial details across the width than the other two options, assuming suitable optics, focus, illumination, and image quality.

3. Ability to Represent a 1.5 mm Defect

To estimate how many pixels represent a defect, divide the physical defect size by the spatial sampling interval.

The formula is:

Pixels across defect = Defect size ÷ Spatial sampling interval

Given:

Minimum defect size = 1.5 mm

3.1 Low Resolution

Spatial sampling interval = 0.9375 mm/pixel

Pixels across defect = 1.5 mm ÷ 0.9375 mm/pixel

= **1.6 pixels**

A 1.5 mm defect would span approximately 1.6 pixels across its width.

This is a very limited representation. The defect may influence one or two pixels, depending on its exact position relative to the pixel grid. Its shape and boundaries may not be represented reliably, and a low-contrast defect may be difficult to distinguish from the fabric background.

Assessment: Low resolution is not recommended for reliable detection of the smallest required defects.

3.2 Medium Resolution

Spatial sampling interval = 0.4902 mm/pixel

Pixels across defect = 1.5 mm ÷ 0.4902 mm/pixel

≈ **3.06 pixels**

A 1.5 mm defect would span approximately 3 pixels across its width.

This is better than the low-resolution option and may be sufficient for detecting some high-contrast defects under favourable conditions. However, approximately three pixels across a defect may provide limited information about its shape and boundaries. Detection of low-contrast oil stains may remain challenging.

Assessment: Medium resolution is a possible compromise, but it requires careful testing to establish whether it can reliably detect all required defect types.

3.3 High Resolution

Spatial sampling interval = 0.2930 mm/pixel

Pixels across defect = 1.5 mm ÷ 0.2930 mm/pixel

≈ **5.12 pixels**

A 1.5 mm defect would span approximately 5 pixels across its width.

This provides a better representation of the defect and its boundaries. It gives the image-processing system more spatial information for distinguishing holes, broken threads, and other small defects.

Assessment: High resolution provides the strongest spatial sampling of the three options and is the preferred candidate when reliable detection of 1.5 mm defects is the priority.

3.4 Defect Representation Summary

Resolution	Sampling interval	Approximate pixels across a 1.5 mm defect	Assessment
Low	0.9375 mm/pixel	1.6 pixels	Inadequate for reliable detection as a general design choice
Medium	0.4902 mm/pixel	3.06 pixels	Potentially usable, but requires validation
High	0.2930 mm/pixel	5.12 pixels	Best candidate among the three for representing small defects

Important: The number of pixels across a defect is a sampling estimate, not a guarantee of successful detection. Actual performance depends on defect contrast, defect orientation, optical resolution, focus, lighting, noise, and the detection algorithm. A defect's physical size alone does not establish its detectability.

4. Comparison of Image Storage, Processing Time, and Defect-Detection Capability

4.1 Image Storage Calculation

Image storage depends on the total number of pixels and the number of bits used to represent each pixel.

For a simple 8-bit grayscale image:

- Each pixel uses 8 bits.
- 8 bits = 1 byte.

Therefore:

Uncompressed grayscale image size in bytes = Total pixels × 1 byte

For an 8-bit RGB image:

- Each pixel contains red, green, and blue values.
- Each channel uses 8 bits.

- Total storage per pixel = 24 bits = 3 bytes.

Therefore:

Uncompressed RGB image size in bytes = Total pixels × 3 bytes

These estimates exclude file headers, metadata, and other overhead. They assume uncompressed image data.

A. Low Resolution

Total pixels = 1,310,720

Grayscale storage:

= 1,310,720 × 1 byte

= 1,310,720 bytes

≈ **1.31 MB**

RGB storage:

= 1,310,720 × 3 bytes

= 3,932,160 bytes

≈ **3.93 MB**

B. Medium Resolution

Total pixels = 5,013,504

Grayscale storage:

= 5,013,504 × 1 byte

= 5,013,504 bytes

≈ **5.01 MB**

RGB storage:

= 5,013,504 × 3 bytes

= 15,040,512 bytes

≈ **15.04 MB**

C. High Resolution

Total pixels = 12,288,000

Grayscale storage:

= 12,288,000 × 1 byte

= 12,288,000 bytes

≈ **12.29 MB**

RGB storage:

= 12,288,000 × 3 bytes

= 36,864,000 bytes

≈ **36.86 MB**

The estimates use decimal megabytes, where 1 MB = 1,000,000 bytes.

4.2 Storage Comparison

Resolution	Total pixels	8-bit grayscale image	8-bit-per-channel RGB image
Low	1,310,720	1.31 MB	3.93 MB
Medium	5,013,504	5.01 MB	15.04 MB
High	12,288,000	12.29 MB	36.86 MB

High-resolution images require substantially more storage than low-resolution images. For example, one uncompressed high-resolution RGB image requires approximately 9.38 times the storage of one low-resolution RGB image.

If images are continuously captured, the storage requirement will depend on the capture rate, the duration of inspection, the number of images retained, and the compression method. Therefore, the table represents storage per image rather than total production-line storage.

4.3 Processing Time

Processing time generally increases with the number of pixels because more image data must be acquired, transferred, and analysed.

For a simplified illustration, assume that a particular algorithm requires the same average amount of computation per pixel at every resolution. Its total pixel-processing workload would then scale approximately with the pixel count.

The relative pixel-processing workloads are:

Low resolution: 1,310,720 pixels

Medium-to-low workload ratio:

= 5,013,504 ÷ 1,310,720

≈ **3.82 times**

High-to-low workload ratio:

= 12,288,000 ÷ 1,310,720

≈ **9.38 times**

High-to-medium workload ratio:

= 12,288,000 ÷ 5,013,504

≈ **2.45 times**

Thus, if the same per-pixel algorithm is used, the high-resolution image may require approximately 9.38 times the pixel-processing work of the low-resolution image. This is a relative estimate, not a measured execution time. Actual processing time depends on the algorithm, implementation, hardware, memory bandwidth, image-transfer overhead, and whether processing can be performed in parallel.

The inspection system has a decision deadline of 0.5 seconds. Therefore, the selected resolution must be tested on the actual processing computer to ensure that image acquisition, processing, and decision generation meet this deadline.

4.4 Defect-Detection Capability

Low resolution:

- Has the lowest processing and storage requirements.
- Provides limited spatial detail.
- A 1.5 mm defect spans only about 1.6 pixels.
- May miss small defects or provide inaccurate defect boundaries.
- Is not recommended when reliable detection of the smallest defects is required.

Medium resolution:

- Provides more detail than low resolution.
- A 1.5 mm defect spans about 3 pixels.
- Offers a useful balance between image detail and computational workload.
- May be sufficient for some defects, but low-contrast or irregular defects need experimental validation.

High resolution:

- Provides the greatest spatial detail among the three options.
- A 1.5 mm defect spans about 5.12 pixels.
- Offers better potential for identifying defect boundaries and small structural changes.
- Requires more memory, data transfer, and processing.
- Must be benchmarked to confirm that the 0.5-second deadline can be achieved.

5. Selection of a Resolution for the Available Computing Resources

The inspection computer has a four-core CPU and 16 GB RAM. The system must inspect moving fabric, detect defects as small as 1.5 mm, and produce a decision within 0.5 seconds.

The selection must balance spatial sampling, image-processing performance, and the need for reliable defect detection.

5.1 Recommended Initial Choice: High Resolution (4096 × 3000 Pixels)

The high-resolution option is the preferred initial design candidate because it provides approximately 5.12 pixels across a 1.5 mm defect when the 4096-pixel image width covers the 1200 mm field of view.

This sampling level offers more spatial information than the low- and medium-resolution alternatives. It is therefore the strongest of the three candidates for representing small holes, broken threads, and other defects.

The main advantage is improved defect visibility. The main disadvantage is the greater computational workload. The high-resolution option should be selected only if benchmarking confirms that the system can meet the 0.5-second decision requirement at the actual acquisition rate.

5.2 Why Not Select Low Resolution?

The low-resolution option represents a 1.5 mm defect using approximately 1.6 pixels across its width. This provides insufficient spatial detail for a robust general-purpose inspection system, especially when defects have low contrast or irregular boundaries.

Although it reduces processing time and storage, this saving is not worthwhile if the system fails to detect defects that the factory needs to identify.

5.3 Why Medium Resolution Is a Backup Option

The medium-resolution option represents a 1.5 mm defect using approximately three pixels across its width. It is a reasonable fallback if high-resolution processing cannot meet the decision deadline on the available computer.

However, the decision to use medium resolution should be based on tests involving representative fabric samples and known defects. If medium resolution cannot meet the required detection performance, the system should retain high resolution and optimise the processing pipeline or acquisition strategy.

5.4 Measures to Balance Image Quality and Processing Time

To make the selected resolution practical, the following measures can be considered:

1. **Efficient image processing:** Use optimised OpenCV operations and avoid unnecessary conversions or repeated processing of the same image.
2. **Region-of-interest processing:** Analyse relevant fabric regions in greater detail when the inspection design allows it, while ensuring that no required inspection area is omitted.
3. **Grayscale processing where suitable:** Use grayscale images for structural defects when colour information is unnecessary. Retain colour or alternative colour-space information when needed to detect stains and colour variations.
4. **Efficient memory management:** Avoid storing unnecessary copies of large images and process images in manageable sections where appropriate.
5. **Performance benchmarking:** Measure the time required for image acquisition, transfer, processing, and decision generation on the actual computer.
6. **Detection validation:** Test each candidate resolution using known 1.5 mm defects, low-contrast stains, broken threads, and holes. Record detection rates and false alarms rather than assuming that higher resolution automatically guarantees success.

Conclusion

The total pixel counts are **1,310,720 pixels** for low resolution, **5,013,504 pixels** for medium resolution, and **12,288,000 pixels** for high resolution. Across the 1200 mm field of view, their approximate spatial sampling intervals are **0.9375 mm/pixel**, **0.4902 mm/pixel**, and **0.2930 mm/pixel**, respectively.

A 1.5 mm defect spans approximately 1.6 pixels at low resolution, 3.06 pixels at m

TASK 7: QUANTIZATION ANALYSIS

Marks: 4

Introduction

Quantization is the process of mapping the intensity values captured by an image sensor to a finite set of digital intensity levels. It determines how accurately differences in brightness can be represented in a digital image. The number of available intensity levels depends on the number of bits used to represent each pixel.

In Sri Lakshmi Textile Industries, the computer vision system must detect fabric defects, including low-contrast oil stains. These stains may differ from the surrounding fabric by only **5–8 grayscale intensity levels**. The lighting conditions may vary between approximately 500 and 3000 lux, so the inspection system must preserve subtle intensity differences while maintaining manageable storage and processing requirements.

Three quantization options are considered: 8-bit quantization, 12-bit quantization, and reduced 4–6-bit quantization.

1. Calculation of the Number of Intensity Levels

The number of intensity levels available in a digital image is calculated using:

Number of intensity levels = 2^n

where *n* is the number of bits used to represent each pixel.

1.1 8-Bit Quantization

Given:

Number of bits, n = 8

Calculation:

Number of intensity levels = 2^8

= 2 × 2 × 2 × 2 × 2 × 2 × 2 × 2

= **256 intensity levels**

The intensity values range from 0 to 255.

- 0 represents black.
- 255 represents white.
- Values between 0 and 255 represent different shades of grey in a grayscale image.

Result: An 8-bit grayscale image provides 256 intensity levels.

1.2 12-Bit Quantization

Given:

Number of bits, n = 12

Calculation:

Number of intensity levels = 2^{12}

= 4096 intensity levels

The intensity values range from 0 to 4095.

Compared with 8-bit quantization, 12-bit quantization provides 16 times as many intensity levels:

4096 ÷ 256 = **16**

This allows smaller changes in the captured signal to be represented, provided the camera sensor and imaging pipeline preserve that information.

Result: A 12-bit image provides 4096 intensity levels.

1.3 Reduced 4–6-Bit Quantization

Reduced quantization uses fewer bits to represent each pixel. The available levels are calculated separately for 4, 5, and 6 bits.

For 4-bit quantization:

Number of levels = 2^4

= **16 intensity levels**

Intensity values range from 0 to 15.

For 5-bit quantization:

Number of levels = 2^5

= **32 intensity levels**

Intensity values range from 0 to 31.

For 6-bit quantization:

Number of levels = 2^6

= **64 intensity levels**

Intensity values range from 0 to 63.

1.4 Summary of Intensity Levels

Quantization option	Bits per pixel	Number of intensity levels	Intensity range
Reduced 4-bit	4	16	0–15
Reduced 5-bit	5	32	0–31
Reduced 6-bit	6	64	0–63
Standard 8-bit	8	256	0–255
High 12-bit	12	4096	0–4095

As the bit depth increases, the image can represent more intensity levels. This can help preserve subtle differences between the fabric and a defect.

2. Effect of Quantization on Low-Contrast Oil Stains

Oil stains on fabric may have only a 5–8 grayscale-level difference from the surrounding fabric. These small intensity differences are important because the inspection system must distinguish the stained region from the normal fabric.

Quantization affects the visibility of such stains because it determines how finely intensity values are represented.

2.1 Effect of 8-Bit Quantization

An 8-bit grayscale image has 256 intensity levels, from 0 to 255.

For example, suppose the normal fabric has an intensity of 120 and an oil stain has an intensity of 114.

Intensity difference = 120 − 114

= **6 grayscale levels**

Since 8-bit quantization can represent integer intensity values throughout the 0–255 range, this difference can be preserved if it is present in the sensor data and the image-processing pipeline does not remove or distort it.

An 8-bit image therefore provides sufficient nominal intensity levels for representing a 5–8-level difference. However, the actual ability to distinguish the stain also depends on sensor noise, illumination uniformity, exposure, and image processing.

Advantages:

- Preserves useful grayscale differences for many industrial inspection tasks.
- Requires relatively little storage.
- Is supported by common image-processing libraries.
- Offers a practical balance between image detail and processing requirements.

Limitation: If lighting changes, sensor noise, or uneven illumination produces intensity variations comparable to the stain contrast, the stain may become difficult to detect even with 8-bit quantization.

2.2 Effect of 12-Bit Quantization

A 12-bit image provides 4096 intensity levels. This allows the captured signal to be represented with finer intensity increments than an 8-bit image, assuming the sensor and acquisition system genuinely provide 12-bit data.

If a 12-bit image is converted directly to 8-bit by dividing its intensity values by 16 and rounding, some fine intensity differences may be lost.

For example, consider two 12-bit intensity values:

Normal fabric = 2000

Oil stain = 1994

Difference = 2000 − 1994

= **6 intensity levels**

If both values are divided by 16 and rounded:

Normal fabric: 2000 ÷ 16 = 125

Oil stain: 1994 ÷ 16 ≈ 124.625, rounded to 125

The two values become equal after this conversion, so the original six-level difference is lost.

This example illustrates why reducing high-bit-depth data without suitable scaling can remove subtle contrast information. In a real system, the result depends on the selected conversion method and the distribution of intensity values.

Advantages:

- Preserves finer intensity distinctions in the acquired signal.
- Provides more flexibility for correcting illumination variations and adjusting contrast.
- Can be beneficial when the camera sensor has sufficient dynamic range and the acquisition pipeline preserves the additional precision.

Limitations:

- Requires more storage when the data are stored using more bits per pixel.
- May increase data-transfer and processing demands.
- Does not automatically improve detection if the sensor noise or lighting variation overwhelms the stain contrast.

- May provide limited practical benefit if the camera's effective precision is much lower than 12 bits.

2.3 Effect of Reduced 4–6-Bit Quantization

Reduced quantization provides fewer intensity levels and therefore groups a wider range of original intensities into each quantized level.

If the full 0–255 grayscale range is mapped uniformly to a lower bit depth, the approximate intensity step in the original 8-bit scale is:

Approximate step size = 256 ÷ Number of quantization levels

For 4-bit quantization:

256 ÷ 16 = **16 original 8-bit intensity values per quantization level**

For 5-bit quantization:

256 ÷ 32 = **8 original 8-bit intensity values per quantization level**

For 6-bit quantization:

256 ÷ 64 = **4 original 8-bit intensity values per quantization level**

A stain with a contrast difference of only 5–8 original grayscale levels may therefore be weakened or lost during reduced quantization.

For example, if two 8-bit intensity values differ by 6 levels, they may map to the same 4-bit quantized value because one 4-bit step represents approximately 16 levels of the original range. They may also map to the same 5-bit value because one 5-bit step represents approximately 8 original levels.

Six-bit quantization uses smaller steps, so a 5–8-level difference is more likely to be represented as a change in quantized value. However, the outcome depends on the exact intensity values and the quantization boundaries; a small difference can still disappear.

Advantages:

- Reduces the number of bits needed to represent each pixel.

- Can reduce memory and data-transfer requirements when the data are genuinely stored in a packed lower-bit format.
- May be suitable for applications where fine intensity distinctions are unimportant.

Limitations:

- Removes subtle grayscale information.
- Can make low-contrast oil stains difficult to distinguish from normal fabric.
- Ma

TASK 8: Image Formation Analysis

Marks: 5

39. Explain the image formation process for a camera mounted 700 mm above the fabric.

Image formation is the process by which a camera captures light reflected from an object and converts it into a digital image.

In Sri Lakshmi Textile Industries, a camera is mounted **700 mm above the moving fabric** to capture images for automatic defect detection and fabric-width measurement.

The image formation process involves the following steps:

1. **Illumination:** A suitable light source illuminates the fabric surface.
2. **Reflection:** Light reflects from the fabric, including defective regions such as oil stains, holes, and irregular patterns.
3. **Lens projection:** The camera lens collects the reflected light and projects it onto the image sensor.
4. **Image sensing:** The sensor converts the received light into electrical signals.
5. **Digital conversion:** These signals are converted into digital pixel values.
6. **Image processing:** Computer vision techniques analyse the captured image to identify defects and measure fabric width.

The camera height affects the field of view, image scale, and measurement accuracy. The lens focal length, sensor dimensions, camera orientation, and working distance must be selected appropriately to cover the required fabric width.

40. Discuss perspective distortion and its effect on fabric-width measurement.

Perspective distortion occurs when an object's appearance or dimensions vary across an image because of the camera's viewing geometry.

If the camera is tilted or the fabric is not parallel to the image sensor, different parts of the fabric may appear at different scales. Consequently, the measured fabric width may differ from its actual physical width.

For example, if the camera is not positioned perpendicular to the fabric, one edge may appear closer to the camera than the other. This can cause errors in estimating the distance between the fabric edges.

Perspective distortion can affect:

- Fabric-width measurement.
- Defect size estimation.
- Defect position identification.
- Consistency of measurements across the fabric surface.

To reduce these errors, the camera should be mounted perpendicular to the fabric, with the optical axis directed towards the inspection region. Calibration and geometric correction should then be used to compensate for remaining distortion.

41. Explain how camera calibration can improve measurement accuracy.

Camera calibration is the process of determining the camera's geometric parameters so that image coordinates can be related more accurately to real-world measurements.

Calibration identifies parameters such as focal length, principal point, and lens distortion. These parameters help correct image distortions and improve dimensional measurements.

The calibration process involves:

1. Capturing images of a calibration target with known dimensions.
2. Estimating the camera's intrinsic parameters and lens distortion.
3. Determining the camera's orientation and position relative to the fabric plane.
4. Applying geometric correction to reduce distortion.

5. Converting pixel distances into physical measurements using a validated image-to-world mapping.
6. Verifying the measurements against a known reference.

For example, if the calibrated image scale is 0.3 mm per pixel, a distance of 100 pixels corresponds to approximately 30 mm, provided the scale is valid in that region and direction.

Calibration can improve measurement accuracy, but it does not guarantee the required accuracy by itself. Camera stability, lighting, lens quality, fabric movement, edge-detection accuracy, and validation results must also be considered.

42. Analyse how lighting variation from 500 to 3,000 lux can affect defect visibility.

Lighting strongly influences image brightness, contrast, and the visibility of fabric defects. In this project, illumination may vary from **500 to 3,000 lux**, which can produce inconsistent image quality if the lighting is not controlled.

Lighting condition	Possible effect on images	Impact on defect detection
500 lux	Images may be too dark, depending on camera settings and fabric reflectance.	Low-contrast defects may become difficult to identify because of noise or insufficient signal.
1,000–2,000 lux	May provide adequate illumination with suitable exposure and lighting geometry.	Defects may be easier to distinguish if brightness is uniform.
3,000 lux	May increase brightness and produce glare or reflections on some fabrics.	Highlights or saturation may hide defects and reduce contrast.

Oil stains in this project may differ from the surrounding fabric by only **5–8 grayscale intensity levels**. Such small differences can be lost when lighting changes significantly or shadows and reflections appear.

To improve defect visibility, the system should use:

- Controlled and uniform illumination.
- A suitable light source and diffuser to reduce shadows and glare.
- Fixed or carefully controlled exposure and camera settings.
- Appropriate image enhancement and contrast analysis.
- Regular verification under realistic operating conditions.

The system should be tested across the expected illumination range to confirm that low-contrast defects remain detectable.

43. Recommend an image formation and illumination arrangement that can achieve the required width accuracy of ±2 mm.

For Sri Lakshmi Textile Industries, the recommended approach is **calibrated imaging with controlled illumination**.

The following arrangement is proposed:

1. **Camera mounting:** Mount the camera approximately 700 mm above the fabric, perpendicular to the fabric surface, using a rigid support to minimise vibration and movement.
2. **Lens selection:** Select a suitable lens and sensor combination that covers the required 1,000 mm fabric width with sufficient resolution and minimal distortion.
3. **Camera calibration:** Calibrate the camera using a target with known dimensions and apply geometric correction to reduce lens and perspective distortion.
4. **Controlled illumination:** Use uniform, diffused lighting to minimise shadows, reflections, and brightness variations across the fabric.
5. **Image acquisition:** Keep camera exposure and other acquisition settings consistent where practical, and synchronise image capture with fabric movement.
6. **Width measurement:** Detect both fabric edges and convert their image coordinates into physical dimensions using the calibrated mapping.
7. **Accuracy validation:** Compare system measurements with a reliable reference measuring instrument at several positions across the fabric width.
8. **Performance verification:** Test the arrangement under different fabric conditions, operating speeds, and illumination levels. Confirm that the processing pipeline meets the required decision time of 0.5 seconds.

Comparison of the candidate approaches

Approach	Advantages	Limitations
Standard perspective imaging without calibration	Simple and inexpensive to implement.	Geometric distortion can cause inaccurate width measurements.

Approach	Advantages	Limitations
Calibrated imaging with geometric correction	Reduces lens and perspective errors and improves dimensional measurement.	Requires calibration, stable mounting, and periodic verification.
Controlled-illumination imaging with suitable calibration	Improves image consistency and measurement reliability by addressing both geometric and lighting effects.	Requires suitable lighting equipment, setup, and validation.

Conclusion

Controlled-illumination imaging combined with camera calibration and geometric correction is the recommended approach. It addresses both geometric measurement errors and lighting-related image variations. However, achieving the required **±2 mm fabric-width accuracy** must be demonstrated through experimental validation; it cannot be guaranteed from the proposed arrangement alone.

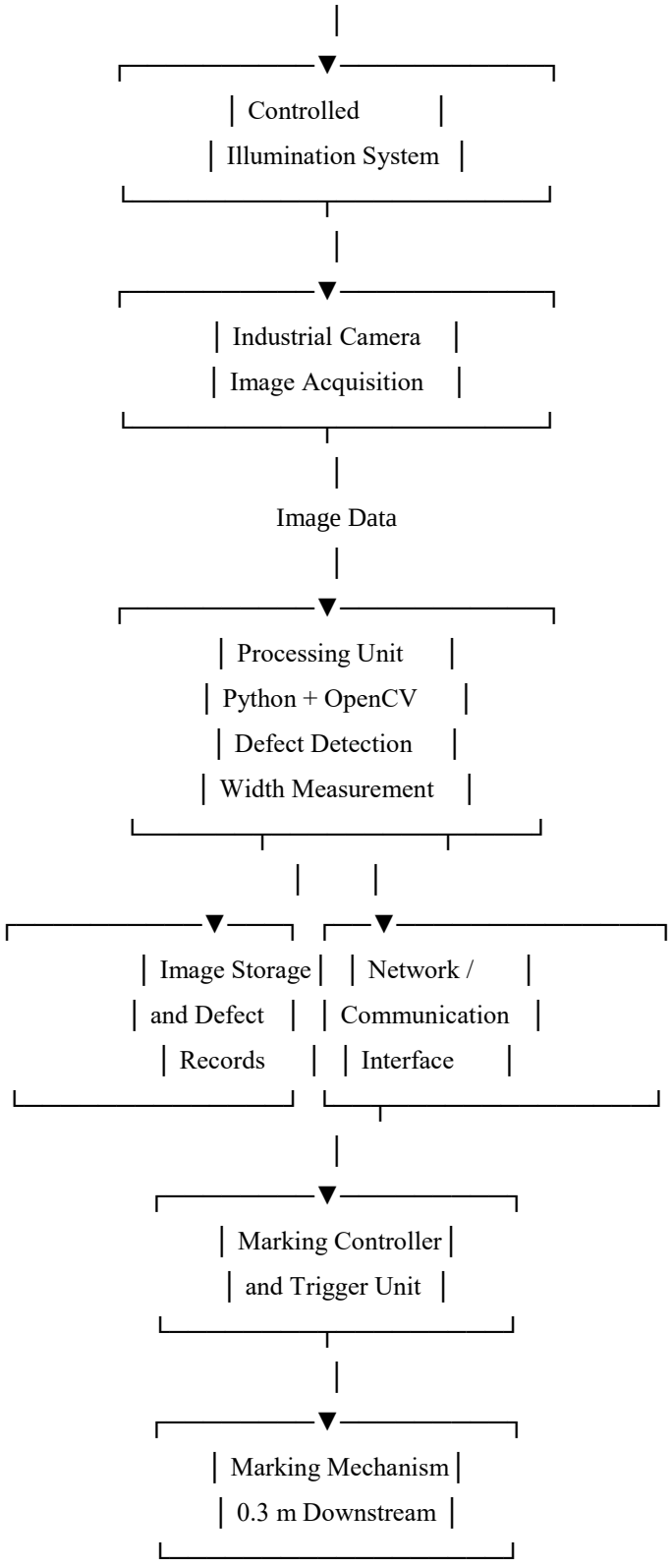
TASK 9: Implementation and Performance Evaluation

Marks: 6

44. Prepare a block diagram showing camera, illumination, processing unit, storage, network and marking mechanism.

Block Diagram





Explanation:

The illumination system provides suitable lighting for the moving fabric. The camera captures fabric images and sends them to the processing unit. Python and OpenCV analyse the images to identify

defects and measure fabric width. Images and defect records are stored for future inspection and reporting. The network interface communicates defect coordinates and timing information to the marking controller, which activates the marking mechanism when the corresponding fabric location reaches the marking point.

The marking controller should use a suitable real-time trigger or encoder-based tracking arrangement to compensate for fabric movement and communication delays.

45. Prepare a comparison table covering estimated cost, resolution, processing time, storage and expected inspection capability.

Three implementation configurations are considered: budget, balanced, and premium.

A. Estimated Configuration Costs

The following costs are **illustrative planning estimates**, not supplier quotations. Actual prices depend on camera specifications, industrial interfaces, lighting, installation, and component availability.

Component	Budget Configuration	Balanced Configuration	Premium Configuration
Camera and lens	₹15,000	₹35,000	₹75,000
Illumination system	₹5,000	₹12,000	₹25,000
Processing hardware	₹30,000	₹45,000	₹1,00,000
Storage and networking	₹10,000	₹12,000	₹20,000
Marking mechanism and controller	₹15,000	₹20,000	₹30,000
Integration and accessories	₹5,000	₹8,000	₹15,000
Total estimated cost	**₹80,000**	**₹1,32,000**	**₹2,65,000**
Within ₹1,50,000 budget?	Yes	Yes	No

These estimates assume one inspection station, one camera, one marking mechanism, and a basic integration effort. Taxes, major production-line modifications, and recurring maintenance costs are excluded.

B. Technical Performance Comparison

Parameter	Budget Configuration	Balanced Configuration	Premium Configuration
Camera resolution	Approximately 1.3 MP	Approximately 5 MP	Approximately 12 MP
Illumination	Basic LED lighting	Controlled, diffused lighting	Advanced industrial lighting
Processing	Basic CPU-based OpenCV	Optimised CPU-based OpenCV	Higher-performance processing hardware
Estimated decision time	0.4–0.8 seconds	0.2–0.5 seconds	0.1–0.3 seconds
Storage	2 TB HDD/SSD allocation	2 TB storage	2 TB or larger storage
Small-defect visibility	Limited	Better, s	

TASK 10: Front-End Adaptation Challenge

Introduction

Sri Lakshmi Textile Industries receives a new export order with more demanding inspection requirements. The system must detect defects as small as 0.4 mm, identify stains visible under ultraviolet (UV) illumination, operate at 1.2 m/s, inspect twice the current production volume, and produce a decision within 0.3 seconds.

However, the existing PC and project budget of ₹1,50,000 cannot be increased.

The existing system was designed for defects of at least 1.5 mm, an initial fabric speed of 0.6 m/s, and a decision deadline of 0.5 seconds. Therefore, selected components and algorithms must be upgraded, and the revised design must be evaluated carefully.

50. Identify which existing system components will become inadequate under the new requirements.

Several components may become inadequate because the new requirements demand higher spatial resolution, faster image acquisition, additional illumination, and shorter processing time.

Existing component	Original requirement	New requirement	Potential limitation
Camera resolution	Detect defects ≥1.5 mm	Detect defects ≥0.4 mm	Existing resolution may not provide enough pixels across small defects.
Camera acquisition rate	Fabric speed 0.6 m/s	Fabric speed 1.2 m/s	Higher acquisition rates are needed to avoid gaps between inspected regions.
Illumination	Visible-light inspection	UV-visible stains must be identified	Existing visible-light illumination may not reveal UV-responsive stains.
Image-processing algorithms	0.5-second decision deadline	0.3-second decision deadline	Existing algorithms may require optimisation to reduce processing time.
Processing hardware	Four-core CPU and 16 GB RAM	Same PC with stricter requirements	Higher-resolution images and increased production volume may exceed processing capacity.

Existing component	Original requirement	New requirement	Potential limitation
Storage	Seven-day image retention	Twice the current production volume	Storage requirements may increase significantly.
Network and controller	Existing marking arrangement	Faster decisions at higher fabric speed	Communication and marking timing must be recalculated.
Optical arrangement	Original defect size and working conditions	0.4 mm defect detection	Lens quality, focus, illumination geometry, and calibration must be re-evaluated.

Conclusion: The camera and illumination are the main candidates for physical upgrades. The acquisition pipeline, processing algorithms, storage policy, and marking controller also require reassessment.

51. Analyse the effect of the increased conveyor speed and smaller defect size on image sampling and acquisition.

The original fabric speed is 0.6 m/s, while the new speed is 1.2 m/s.

A. Effect of Increased Conveyor Speed

The new conveyor speed is:

v=1.2 m/s=1200 mm/sv=1.2\text{ m/s}=1200\text{ mm/s}

The fabric now moves twice as fast as before. Consequently, the camera must capture images more frequently or use an appropriate line-scan acquisition rate to maintain sufficient coverage.

For example, if the maximum permitted movement between successive samples along the fabric is 0.2 mm, the minimum sampling rate is:

f=Fabric speedSampling intervalf=\frac{\text{Fabric speed}}{\text{Sampling interval}}
f=12000.2f=\frac{1200}{0.2} f=6000 samples/s\boxed{f=6000\text{ samples/s}}

This is an illustrative design calculation. The required sampling rate depends on the chosen sampling interval, the camera type, and the defect-detection method.

If the acquisition rate is too low, small defects may fall between sampled regions and remain undetected.

B. Effect of Smaller Defect Size

The minimum defect size decreases from 1.5 mm to 0.4 mm.

The reduction factor is:

1.50.4=3.75\frac{1.5}{0.4}=3.75

Therefore, the new system must detect defects approximately 3.75 times smaller in physical size.

For reliable detection, the image should contain multiple pixels across a defect. As a preliminary design target, assume at least five pixels across a 0.4 mm defect.

The required spatial sampling interval is:

Sampling interval≤0.45\text{Sampling interval}\leq\frac{0.4}{5}
Sampling interval≤0.08 mm/pixel\boxed{\text{Sampling interval}\leq0.08\text{ mm/pixel}}

Across a 1,200 mm horizontal field of view, the required horizontal pixel count would be:

N=12000.08N=\frac{1200}{0.08} N=15,000 pixels\boxed{N=15,000\text{ pixels}}

This calculation shows that a standard 5 MP area-scan camera may not provide enough horizontal pixels if it must cover the entire 1,200 mm field of view in one image.

A higher-resolution camera, multiple cameras, or a suitable line-scan arrangement may be required. The final design must also verify sampling along the fabric's direction of travel.

C. Effect on Motion Blur

At 1.2 m/s, the fabric moves 1.2 mm during a 1 ms exposure.

To limit motion blur to 0.08 mm:

t=0.081200t=\frac{0.08}{1200} t=0.0000667 secondst=0.0000667\text{ seconds}
t≈66.7 microseconds\boxed{t\approx66.7\text{ microseconds}}

Therefore, a short exposure time and sufficient illumination are important. The actual permissible blur should be determined through testing because it depends on defect contrast, orientation, and the detection algorithm.

Conclusion: The higher conveyor speed and smaller defect size require finer spatial sampling, faster acquisition, shorter exposure times, and careful synchronisation.

52. Explain how UV illumination or an additional imaging channel could help identify the new stain type.

Some stains may be difficult to distinguish under ordinary visible illumination. Under suitable ultraviolet illumination, certain materials absorb UV energy and emit visible light through fluorescence, while others may produce different responses from the surrounding fabric.

The response depends on the stain's chemical composition, fabric material, dyes, and illumination wavelength. Therefore, UV-based detection must be validated using representative stained and unstained samples.

Proposed UV Inspection Method

1. Install a suitable UV light source with the required wavelength.
2. Select a camera sensor and optical filter arrangement compatible with the expected stain response.
3. Capture images under controlled UV illumination.
4. Compare the stain region with the surrounding fabric.
5. Apply intensity, colour, or fluorescence-based analysis where appropriate.
6. Record the detected stain's location, dimensions, and confidence.
7. Validate detection performance using different fabric types and stain samples.

Additional Imaging Channel

The system may use two imaging channels:

- **Visible-light channel:** Detects holes, broken threads, and weaving irregularities.
- **UV-responsive channel:** Helps identify stains that produce a distinguishable response under UV illumination.

The channels may use separate cameras or a carefully designed optical arrangement. Separate cameras can simplify channel-specific filtering, but they increase hardware cost and require image alignment and synchronisation.

UV illumination alone does not guarantee stain detection. The selected wavelength, camera sensitivity, filters, exposure, and safety measures must match the actual stain characteristics.

Conclusion: A UV-responsive imaging channel can improve detection of stains that are difficult to distinguish under visible light, provided their optical response is suitable and experimentally verified.

53. Compare the three adaptation strategies in terms of feasibility, accuracy, processing time, cost and implementation risk.

Three strategies are considered.

Strategy 1: Retain the Existing System with Only Software Changes

This strategy keeps the existing camera, lighting, PC, and other hardware. The team optimises image-processing algorithms, improves thresholding and texture analysis, and reduces unnecessary computations.

Advantages:

- Lowest additional cost.
- Minimal hardware changes.
- Shorter implementation period.

Limitations:

- Cannot compensate for insufficient camera resolution.
- Cannot reliably identify UV-responsive stains without suitable illumination and sensing hardware.
- May not support the acquisition rate required at 1.2 m/s.
- Processing optimisation cannot overcome fundamental optical limitations.

Assessment: This approach is inexpensive but unlikely to satisfy all the new requirements.

Strategy 2: Upgrade Selected Existing Components

This strategy retains the existing PC and upgrades selected components, such as the camera, lens, illumination system, and acquisition interface.

The software is also optimised to reduce processing time.

Advantages:

- Better image quality and spatial sampling.
- UV illumination can be added for suitable stains.
- Existing PC and software can be reused.
- Lower cost than replacing the entire system.

Limitations:

- A sufficiently high-resolution camera may be expensive.
- The existing PC may struggle with high-resolution images at the required speed.
- UV hardware and additional optical components increase integration complexity.
- Achieving the 0.3-second deadline and 2% maximum missed-defect rate is uncertain without benchmarking.

Assessment: This is the most practical initial approach under the unchanged budget, provided component selection is based on measured performance and actual quotations.

Strategy 3: Redesign the System Using a Different Acquisition and Processing Architecture

This strategy redesigns the inspection system around an architecture such as line-scan acquisition, encoder-synchronised capture, dedicated processing, and separate visible and UV-responsive imaging channels.

Advantages:

- Better potential for continuous-web inspection at high speed.
- Flexible spatial sampling and acquisition timing.
- Improved integration of visible and UV inspection.
- Greater potential to support demanding inspection requirements.

Limitations:

- Highest integration and testing complexity.
- May require additional cameras, specialised optics, and processing hardware.

- Could exceed the available budget.
- Requires substantial development and validation.

Assessment: This offers the strongest potential technically but may not be financially or computationally feasible under the existing constraints.

Comparison Table

Evaluation factor	Strategy 1: Software only	Strategy 2: Selected upgrades	Strategy 3: Full redesign
Feasibility	Low for all new requirements	Moderate	Technically promising, but resource-constrained
Accuracy	Limited by existing optics and sensor	Potentially high after validation	Potentially highest after validation
Processing time	May improve modestly	May meet the deadline with efficient algorithms	Depends on architecture and available processing hardware
Additional cost	Very low	Moderate	High
UV stain detection	Not feasible without suitable UV sensing hardware	Possible with suitable UV illumination and sensing	Can support a dedicated UV-responsive channel
Implementation risk	High risk of failing requirements	Moderate to high	High integration and cost risk
Suitability for unchanged budget	High financially	Best candidate	Low

54. Recommend the most practical strategy under the unchanged PC and budget constraints.

Recommendation: Strategy 2 — Upgrade selected components while retaining the existing PC.

This strategy offers the best compromise between performance, implementation cost, and reuse of existing resources.

The following changes are recommended:

1. Replace the existing camera if its resolution or acquisition rate is insufficient.

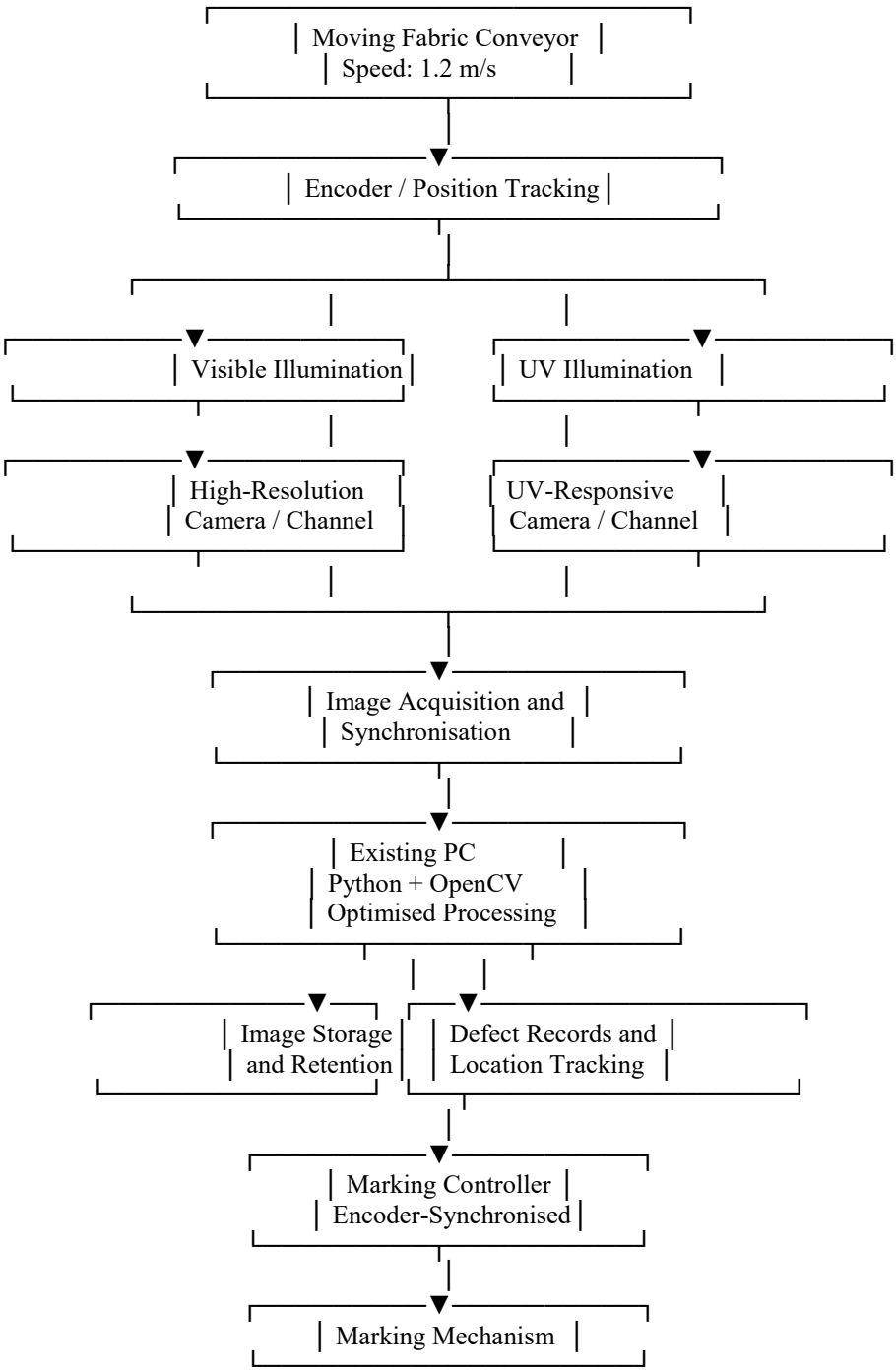
2. Select a suitable lens and optical arrangement to support the required spatial sampling.
3. Add UV illumination and an appropriate camera/filter arrangement for UV-responsive stains.
4. Optimise Python and OpenCV processing to reduce unnecessary operations.
5. Use region-of-interest processing where it does not compromise defect coverage.
6. Reduce image-transfer overhead and avoid unnecessary full-frame copies.
7. Use efficient acquisition, buffering, and encoder synchronisation.
8. Review the storage policy to accommodate twice the production volume.
9. Test the complete system under maximum speed and worst-case image conditions.

However, the final camera and UV arrangement must be chosen only after obtaining quotations and checking compatibility with the existing PC.

If no combination of suitable components fits the budget and meets the measured performance requirements, the company must revise the requirements, reduce the inspected field per camera, or obtain additional resources.

55. Prepare an updated system block diagram and list the changes required.

Updated System Block Diagram



Note: The two camera channels are a proposed design option. If one camera can reliably acquire the required visible and UV-responsive information, a single-camera arrangement may reduce cost. This must be confirmed by testing.

Required Changes

Component	Proposed change	Purpose
Camera	Select a camera or camera arrangement with adequate spatial resolution and acquisition rate	Detect 0.4 mm defects
Lens and mounting	Re-evaluate optics, field of view, focus, and calibration	Improve measurement consistency
Illumination	Add controlled UV illumination and retain suitable visible lighting	Detect UV-responsive stains and ordinary defects
Acquisition	Improve triggering and synchronisation	Maintain continuous coverage at 1.2 m/s
Processing software	Optimise OpenCV operations and image transfer	Target the 0.3-second deadline
Storage	Recalculate capacity and implement a validated retention policy	Support twice the production volume
Position tracking	Use an encoder or equivalent movement tracking	Deliver marking commands at the correct location
Testing	Measure latency, detection rate, and image retention	Verify the requirements objectively

56. Clearly identify any requirements that cannot realistically be achieved without additional resources, and justify your conclusion.

Some requirements may be achievable through careful component selection and software optimisation, but they cannot be guaranteed while keeping both the PC and budget unchanged.

A. Detecting 0.4 mm Defects Across the Full Fabric Width

The previous design considered a 1,200 mm field of view and a target defect size of 1.5 mm. The new 0.4 mm target requires much finer spatial sampling.

At five pixels across a 0.4 mm defect, the approximate sampling interval must be 0.08 mm/pixel. Covering 1,200 mm at that sampling interval requires approximately 15,000 horizontal pixels.

This is a demanding acquisition requirement. A suitable line-scan camera or multiple-camera arrangement may help, but the optical resolution, line rate, exposure, and full-width coverage must be verified.

Conclusion: This requirement may not be realistic with the existing camera if it lacks sufficient resolution or acquisition capability. A camera upgrade or alternative acquisition arrangement is likely to be necessary.

B. Detecting UV-Responsive Stains

The existing visible-light system may not reveal stains that become distinguishable only under UV illumination.

Conclusion: Without suitable UV illumination and a camera/filter arrangement capable of capturing the relevant response, software changes alone cannot reliably satisfy this requirement.

C. Operating at 1.2 m/s with a 0.3-Second Decision Deadline

The new conveyor speed is twice the original speed, while the decision deadline decreases from 0.5 seconds to 0.3 seconds.

The system must acquire images, transfer data, process defects, and produce a decision within the shorter deadline.

Conclusion: This requirement cannot be guaranteed on the existing PC without benchmarking. If processing time exceeds 0.3 seconds under the maximum expected workload, a more efficient acquisition architecture, reduced computational workload, or additional processing resources may be required.

D. Inspecting Twice the Production Volume and Retaining Images for Seven Days

Twice the production volume may increase the total number of images and defect records. Storage needs depend on the capture rate, image resolution, compression, and whether all full-resolution images must be retained.

Conclusion: Seven-day retention may remain feasible with the existing storage if a validated image-management policy is used. If the required data volume exceeds available capacity, additional storage or an approved retention strategy will be necessary.

E. Achieving a Maximum Missed-Defect Rate of 2%

A high-resolution camera and UV illumination can improve the information available to the algorithm, but neither guarantees that no more than 2% of defects will be missed.

The system must be evaluated on representative samples covering the different defect types, fabrics, illumination conditions, and operating speeds.

Conclusion: The 2% missed-defect target cannot be guaranteed without experimental validation. Additional labelled samples, algorithm improvements, and possibly additional sensing hardware may be needed.

Final Conclusion

Under the unchanged PC and ₹1,50,000 budget, **upgrading selected components while retaining the existing PC is the most practical initial strategy**. The company should prioritise suitable camera resolution, UV-responsive inspection, controlled illumination, efficient acquisition, and software optimisation.

However, the combined requirements of 0.4 mm defect detection, 1.2 m/s operation, a 0.3-second decision deadline, twice the production volume, and a maximum 2% missed-defect rate may exceed the existing system's capabilities. If testing demonstrates that these targets cannot be met within the available resources, the company must either obtain additional resources or negotiate a technically justified adjustment to the requirements.

APPENDIX A – PYTHON / OPENCV IMPLEMENTATION OUTLINE

A.1 Objective

The objective is to develop a computer vision system using Python, OpenCV, and NumPy to detect fabric defects, measure fabric width, record defect information, and support defect marking.

A.2 System Workflow

The proposed workflow is:

Image Acquisition → Preprocessing → Defect Detection → Classification → Coordinate Calculation → Data Storage → Defect Marking

A.3 Defect Detection Methods

Defect	Proposed technique
Holes	Thresholding and contour detection
Broken threads	Edge detection and line analysis
Oil stains	Intensity, contrast, and colour analysis
Weaving irregularities	Texture and pattern analysis

A.4 Width Measurement

The physical width is estimated using calibrated image coordinates.

Measured Width = Pixel Distance × Scale (mm/pixel)

Camera calibration is necessary to improve measurement accuracy.

A.5 Defect Recording

Each defect record should contain the defect ID, category, timestamp, coordinates, estimated size, and image reference.

A.6 Performance Requirements

The proposed system should target:

- Minimum defect size: 1.5 mm.
- Width measurement accuracy: ±2 mm.
- Decision time: ≤0.5 seconds.
- Missed-defect rate: ≤2%.

These targets require experimental verification.

A.7 Conclusion

Python and OpenCV provide a suitable starting point for developing the prototype. Actual detection accuracy and processing time must be measured during testing.

APPENDIX B – DEFECT FEATURES AND VALIDATION PLAN

B.1 Defect Feature Analysis

Defect	Main features	Proposed detection method
Holes	Open regions and irregular boundaries	Thresholding and contour analysis
Broken threads	Interrupted yarn lines	Edge and line analysis
Oil stains	Intensity and colour differences	Contrast analysis and segmentation
Weaving irregularities	Abnormal texture and yarn arrangement	Texture analysis

Oil stains may differ by only 5–8 grayscale levels, making illumination and contrast important.

B.2 Validation Plan

The system should be tested using labelled images of normal fabric and all four defect categories.

Parameter	Target
Fabric speed	0.6 m/s
Minimum defect size	1.5 mm
Illumination	500–3000 lux
Width measurement error	≤2 mm

Parameter	Target
Decision time	≤0.5 seconds
Missed-defect rate	≤2%
Data retention	Seven days

B.3 Evaluation Metrics

- **Precision:** Percentage of detected defects that are genuine.
- **Recall:** Percentage of actual defects detected.
- **Missed-defect rate:** Percentage of actual defects missed.
- **Processing time:** Time required to produce a defect decision.

Missed-Defect Rate = FN / (TP + FN) × 100%

Here, FN represents false negatives and TP represents true positives.

B.4 Limitations

Normal fabric patterns may resemble defects. Low-contrast stains, lighting variations, and fabric movement may affect detection performance.

B.5 Conclusion

Validation must establish whether the system meets its detection, measurement, processing-time, and storage requirements. Results should be based on actual experiments.

APPENDIX C – CALIBRATION AND COMMISSIONING CHECKLIST

C.1 Objective

Calibration establishes the relationship between image pixels and physical dimensions. Commissioning verifies that the complete inspection system operates as intended.

C.2 Calibration Checklist

Checkpoint	Verification
Camera installation	Confirm mounting and alignment
Camera height	Approximately 700 mm, subject to confirmation
Field of view	Verify 1200 × 400 mm
Image focus	Check image sharpness
Illumination	Check brightness and uniformity
Camera calibration	Use a reference target
Width measurement	Compare with a reference measurement
Software	Verify Python and OpenCV operation

C.3 Calibration Formula

Scale = Physical Distance / Pixel Distance

For example, if 1200 mm corresponds to 4000 pixels:

Scale = 1200 / 4000 = **0.3 mm/pixel**

This is a nominal scale; actual accuracy must be verified.

C.4 Performance Verification

The system should be tested for:

- Detection of all four defect categories.
- Width measurement error ≤2 mm.
- Decision time ≤0.5 seconds.
- Missed-defect rate ≤2%.
- Correct defect-location tracking and marking.
- Seven-day data retention under the selected recording policy.

C.5 Commissioning Status

Each checkpoint should be marked **Pass, Fail, or Pending** based on recorded test results. Requirements must not be marked as passed without evidence.

C.6 Conclusion

Calibration and commissioning help establish reliable image measurements and system operation. The system should enter a controlled trial only after the relevant checks have been completed and documented.

APPENDIX D – STORAGE, REFERENCES AND FINAL DECISION

D.1 Storage Calculation

For an uncompressed 5-megapixel grayscale image:

Image size ≈ **5.01 MB**

For continuous recording at one image per second for seven days:

Total storage = Image Size × Images per Second × Recording Duration

Total storage ≈ 5.01 × 604,800 MB

Total ≈ 3.03 TB

Therefore, 2 TB is insufficient for this continuous, uncompressed recording scenario.

Possible solutions include defect-based image storage, compression, reduced archival frequency, or additional storage. The chosen approach must preserve sufficient evidence for quality control.

D.2 Technical References

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9. Steger, C., Ulrich, M., and Wiedemann, C., *Machine Vision Algorithms and Applications*, Wiley-VCH.
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D.3 Final Engineering Decision

The proposed system can be developed as a prototype using Python, OpenCV, NumPy, and a suitable camera.

However, the following requirements must be verified experimentally:

- Defect detection and classification.
- Missed-defect rate ≤2%.
- Width measurement accuracy of ±2 mm.
- Decision time ≤0.5 seconds.
- Reliable defect marking.
- Seven-day storage retention.

D.4 Final Recommendation

Proceed with prototype development and controlled validation. Calibrate the camera, prepare labelled defect images, implement the inspection workflow, measure actual performance, and resolve storage limitations before considering industrial deployment.

D.5 Conclusion

The proposed system offers a structured approach to automated fabric defect inspection. Its industrial suitability depends on successful validation of detection accuracy, measurement precision, processing time, and storage requirements.